Dependency-Aware Multi-Level Feature-Guided U-Net with Audit-Gated Causal Memory for Long Text-to-Video Generation

by

ROHAN SANJAY POL
(Roll No: M240006CS)

DISSERTATION

Submitted for the award of the Degree of

Master of Technology

in

Artificial Intelligence and Machine Learning



Department of Computer Science and Engineering

NATIONAL INSTITUTE OF TECHNOLOGY SIKKIM

June, 2026

NATIONAL INSTITUTE OF TECHNOLOGY SIKKIM
Ravangla Campus
Barfung Block, Ravangla Sub Division, South Sikkim-737139
(AN INSTITUTE OF NATIONAL IMPORTANCE, GOVT. OF INDIA)

Certificate by Supervisor

This is to certify that the project report entitled "**Dependency-Aware Multi-Level Feature-Guided U-Net with Audit-Gated Causal Memory for Long Text-to-Video Generation**" is being submitted by Rohan Sanjay Pol (Roll No: M240006CS), a student in the Department of Computer Science and Engineering, National Institute of Technology Sikkim, for the award of the degree of Master of Technology (M.Tech). It is an original work carried out by him under my supervision and guidance. The results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.

Dr. Krishna Kumar
(Assistant Professor)
Department of CSE
NIT Sikkim

NATIONAL INSTITUTE OF TECHNOLOGY SIKKIM
Ravangla Campus
Barfung Block, Ravangla Sub Division, South Sikkim-737139
(AN INSTITUTE OF NATIONAL IMPORTANCE, GOVT. OF INDIA)

Certificate by Student

I hereby declare that the work presented in the report entitled "Dependency-Aware Multi-Level Feature-Guided U-Net with Audit-Gated Causal Memory for Long Text-to-Video Generation"is a bonafide record of the work done by me under the supervision of Dr. Krishna Kumar, Assistant Professor, Department of Computer Science and Engineering and that no part thereof has been presented for the award of any other degree.

- I have followed the guidelines provided by the institute in writing the report.
- The report does not contain any classified information.
- Whenever, I have used materials from other sources, I have given due credits to those by citing them in the text of the report and giving the details in the references.
- Whenever, I have quoted written materials from other sources, I have put them under quotation marks and given due credits to the sources by citing them in the text of the report and giving their details in the references.

Dated: 02 June, 2026
Place: Ravangla

Mr. Rohan Sanjay Pol
Roll No.: M240006CS

©NATIONAL INSTITUTE OF TECHNOLOGY SIKKIM, COPYRIGHT
©2026 ALL RIGHTS RESERVED

Acknowledgments

I express my deepest gratitude to my supervisor, **Dr. Krishna Kumar**, Assistant Professor, Department of Computer Science and Engineering, National Institute of Technology Sikkim, for his unwavering guidance, continuous support, and constant encouragement throughout the course of this work. His invaluable expertise, insightful suggestions, and keen observations were instrumental in the successful development and completion of this research. The knowledge and skills acquired under his mentorship will greatly contribute to my future academic and professional endeavors.

I would also like to express my sincere gratitude to **Prof. M. C. Govil**, Director, National Institute of Technology Sikkim, and **Dr. Taraknath Kundu**, Dean, Research & Consultancy, National Institute of Technology Sikkim, for their patronage and for providing the necessary academic and research environment to carry out this work successfully.

I extend my heartfelt thanks to **Dr. Sangram Ray**, Head of the Department, Computer Science And Engineering, for his valuable support, encouragement, and motivation throughout my postgraduate studies.

I am also thankful to all the faculty members of the Department of Computer Science and Engineering for imparting strong fundamental knowledge in Computer Science , which proved highly beneficial during the course of this research work.

Finally, I would like to express my sincere gratitude to my family and friends for their continuous encouragement, patience, and moral support throughout the completion of this project.

Date: June 6, 2026
Place: Ravangla, Sikkim **Rohan Sanjay Pol**

Contents

List of Figures vii

List of Tables viii

List of Acronyms ix

List of Notations xi

Abstract xviii

1 Introduction 1

1.1 Background 1
1.2 Motivation 4
1.3 Problem Statement 6
1.4 Objectives 6
1.5 Problem Formulation 7
1.6 Chapter Summary 9

2 Literature Review 10

2.1 Tools and Technologies Used 10
2.2 Related Work 13
2.2.1 Diffusion and Latent Diffusion Models 13
2.2.2 Text-to-Video Generation 14
2.2.3 Long-Form and Multi-Scene Video Generation 15
2.2.4 LLM-Based Video Planning and Scene Decomposition 15
2.2.5 Visual Consistency in Generated Videos 16
2.2.6 Memory and Retrieval-Based Generation 16
2.2.7 U-Net Conditioning and Adapter-Based Diffusion Control 17
2.2.8 Video Generation Evaluation and Repair 18
2.2.9 Summary of Research Gap 18
2.3 Chapter Summary 19

3 Proposed Works **20**
3.1 Proposed Works . . . 20
3.1.1 Overview of the Proposed DAMF-CMG Framework . . . 20
3.1.2 Problem Formulation . . . 23
3.1.3 Transition-Aware Scene State Construction . . . 24
3.1.4 Repair-Aware Causal Scene Dependency Detection . . . 25
3.1.5 Constraint-Aware Scene Packet Builder . . . 27
3.1.6 Dependency-Aware Multi-Level Feature Extraction . . . 29
3.1.7 Audit-Gated Causal Memory Bank . . . 30
3.1.8 Dependency-Aware Memory Retrieval and Continuity Package Construction . . . 32
3.1.9 DAMF-U-Net: Dependency-Aware Memory-Guided Denoising . 33
3.1.10 Timestep-Aware and Layer-Aware Memory Control . . . 36
3.1.11 Scene Audit, Selective Repair, and Downstream Regeneration . . 37
3.1.12 Training Objective and Optimization . . . 38
3.1.13 Overall Algorithm and Methodology Summary . . . 40
3.2 Experimental Setup . . . 42
3.2.1 Dataset and Prompt Benchmark . . . 42
3.2.2 Baseline Methods . . . 42
3.2.3 Implementation Details . . . 43
3.2.4 Evaluation Metrics . . . 44
3.2.5 Ablation Protocol . . . 45
3.2.6 Human Evaluation Protocol . . . 45
3.3 Results and Discussion . . . 46
3.3.1 Quantitative Comparison with Baselines . . . 46
3.3.2 Long-Video Consistency Analysis . . . 48
3.3.3 Qualitative Results Using Generated Video Frames . . . 49
3.3.4 Memory and Repair Effectiveness . . . 50
3.3.5 Discussion of DAMF-U-Net Behavior . . . 50
3.3.6 Failure Case Analysis . . . 51
3.3.7 Summary of Findings . . . 52
3.4 Chapter Summary . . . 52

4 Conclusions and Future Works **53**
4.1 Limitations . . . 53
4.2 Conclusion and Future Work . . . 54

Bibliography **60**

List of Figures

3.1 Overall DAMF-CMG framework. The proposed framework converts a long text prompt into transition-aware scene states, constructs a repair-aware causal dependency graph, and builds constraint-aware scene packets for each scene. Dependency-relevant committed memories are retrieved from the audit-gated causal memory bank and injected into DAMF-U-Net for memory-guided denoising. Generated scenes are audited before memory commitment. Passed scenes update the committed memory bank and contribute to the final long video, whereas failed scenes trigger selective repair and regeneration of only causally affected downstream scenes. 22
3.2 Repair-aware dependency graph and memory flow. This figure should show source scenes, target scenes, typed dependency edges, dependency scores, memory types, committed memory, and repair impact. 27
3.3 DAMF-U-Net architecture with dependency-aware memory conditioning. 33
3.4 Qualitative results of scene-wise long text-to-video generation. 46

List of Tables

2.1 Tools and technologies used in DAMF-CMG. 12

3.1 Quantitative comparison with baseline methods. 48
3.2 Long-Video Consistency Results. 49
3.3 Effect of memory type on consistency dimensions. 51

List of Acronyms

Acronym	Full Form / Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CLIP	Contrastive Language–Image Pretraining
CLIPScore	Contrastive Language–Image Pretraining Score
CUDA	Compute Unified Device Architecture
DAMF-CMG	Dependency-Aware Multi-Level Feature-Guided U-Net with Audit-Gated Causal Memory
DAMF-U-Net	Dependency-Aware Multi-Level Feature-Guided U-Net
DDPM	Denoising Diffusion Probabilistic Model
DINO	Self-Distillation with No Labels
DINOv2	Second version of DINO visual representation model
DL	Deep Learning
DRR	Downstream Regeneration Ratio
EvalCrafter	Evaluation benchmark for text-to-video generation
FFmpeg	Fast Forward MPEG multimedia processing tool
FiLM	Feature-wise Linear Modulation
FPS	Frames Per Second
FVD	FrÃ©chet Video Distance
GPU	Graphics Processing Unit
GroundingDINO	Grounding DINO phrase-grounded detection model
IP-Adapter	Image Prompt Adapter
LDM	Latent Diffusion Model
LLM	Large Language Model
mAP	Mean Average Precision
MCR	Memory Contamination Rate

Continued on next page

List of Acronyms

Acronym	Full Form / Meaning
ML	Machine Learning
MLP	Multi-Layer Perceptron
MovieBench	Movie-level video generation benchmark
MSR-VTT	Microsoft Research Video to Text dataset
NLP	Natural Language Processing
OCR	Optical Character Recognition
RSR	Repair Success Rate
SAM	Segment Anything Model
SAM2	Segment Anything Model 2
SBERT	Sentence-BERT sentence embedding model
T2I-Adapter	Text-to-Image Adapter
T2V	Text-to-Video
T2V-CompBench	Text-to-Video Compositional Benchmark
U-Net	U-shaped Network
UNet	U-shaped Neural Network Architecture
VAE	Variational Autoencoder
VATEX	Video-And-Text dataset
VBench	Video Generation Benchmark
VFID	Video FrÃ©chet Inception Distance
W&B	Weights and Biases experiment tracking

List of Notations

Symbol	Description
Prompt, scene, and video notation	
P	Long input text prompt.
w_n	n-th token or word in the input prompt.
$S = \{s_1, s_2, \ldots, s_N\}$	Ordered sequence of structured scene states.
s_i	Current or target scene.
s_j	Source or earlier scene, where $j < i$.
N	Total number of scenes in the prompt.
V_i	Generated video segment for scene s_i.
$\bar{V}_i$	Accepted or repaired version of generated scene video V_i.
V	Final generated long video.
$x_{i,f}$	f-th frame of scene video V_i.
F_i	Number of frames in scene video V_i.
F	Fixed number of frames per scene when all scenes use the same length.
Scene state notation	
d_i	Natural-language description of scene s_i.
E_i	Entities or characters appearing in scene s_i.
L_i	Location or background of scene s_i.
O_i	Objects or props in scene s_i.
A_i	Main action in scene s_i.
T_i	Temporal state of scene s_i.
C_i	Continuity constraints of scene s_i.
$\mathcal{R}_i^{req}$	Reference requirement for scene s_i.
Δ_i	Transition state from scene s_{i-1} to scene s_i.
$\phi(s_{i-1}, s_i)$	Transition extraction function between adjacent scenes.
Dependency graph notation	

Continued on next page

Symbol	Description
$G_d = (V_d, E_d)$	Repair-aware causal scene dependency graph.
V_d	Set of scene nodes in dependency graph G_d.
E_d	Set of directed dependency edges in dependency graph G_d.
$e_{j\to i}$	Directed dependency edge from source scene s_j to target scene s_i.
$D_{j\to i}$	Dependency score from scene s_j to scene s_i.
$\mathbf{z}_{j\to i}$	Typed dependency vector from source scene s_j to target scene s_i.
$z^k_{j\to i}$	Binary indicator showing whether dependency type k is active.
$\rho_{j\to i}$	Dependency strength from scene s_j to scene s_i.
$\bar{\rho}_{j\to i}$	Normalized dependency strength.
$\kappa_{j\to i}$	Repair impact score from source scene s_j to target scene s_i.
θ_d	Threshold for adding a dependency edge.
θ_1, θ_2	Thresholds for classifying independent, soft, and hard dependencies.
γ_1, γ_2	Thresholds used for downstream recovery decisions.
Scene packet notation	
P_i	Constraint-aware scene packet for scene s_i.
$\Psi(\cdot)$	Scene packet construction function.
Ω_i	Observation state of scene s_i.
$\mathcal{I}_i$	Invariant constraints that should remain stable in scene s_i.
$\mathcal{A}_i^{chg}$	Allowed-change constraints for scene s_i.
$\mathbf{q}_i$	Continuity priority vector for scene s_i.
q_i^k	Continuity priority of memory type k in scene s_i.
$\mathbf{r}_i$	Memory requirement vector for scene s_i.
r_i^k	Binary indicator showing whether memory type k is required.
$\mathbf{g}_i$	DAMF-U-Net memory conditioning strength vector.
g_i^k	Conditioning strength of memory type k for scene s_i.
$\mathcal{G}_i$	Generation contract for scene s_i.

Continued on next page

Symbol	Description
Q_i	Audit targets for scene s_i.
$\mathcal{U}_i$	Memory update policy for scene s_i.
$\mathcal{H}_i$	Repair policy hints for scene s_i.
Γ_i	Continuity package assembled from retrieved memory and scene constraints.
Memory and retrieval notation	
$\mathcal{B}_{i-1}$	Committed memory bank before generating scene s_i.
$\mathcal{B}_i$	Updated memory bank after processing scene s_i.
$\mathcal{M}_i$	Extracted memory set from generated scene video V_i.
μ_i^k	Memory item of type k extracted from scene s_i.
m_i^k	Retrieved memory feature of type k used for target scene s_i.
$\mathcal{R}_i$	Retrieved memory set for scene s_i.
$\mathcal{F}_{extract}(\cdot)$	Dependency-aware feature extraction function.
Retrieve$(\cdot)$	Function that selects valid and dependency-compatible memories.
Conf(m_i^k)	Confidence score of retrieved memory feature m_i^k.
Conf(μ_i^k)	Confidence score of memory item μ_i^k.
$\mathcal{K}$	Set of memory types.
k	Memory type index.
$k \in \{\text{id, app, loc, prop, style, motion, trans}\}$	Memory types representing identity, appearance, location, prop, style, motion, and transition.
Feature extraction notation	
X_i^{ref}	Selected reference frames or clips for scene s_i.
x_i^{id}	Identity reference frame for scene s_i.
x_i^{loc}	Location reference frame for scene s_i.
x_i^{prop}	Prop reference frame for scene s_i.
x_i^{style}	Style reference frame for scene s_i.
x_i^{motion}	Motion reference clip or frame set for scene s_i.
x_i^{last}	Last frame used for transition continuity.

Continued on next page

Symbol	Description
b_e^f	Bounding box of entity e in frame f.
M_e^f	Segmentation mask of entity e in frame f.
q_e	Phrase query used to localize entity e.
m_{char}^{low}	Low-level character memory, such as color or texture cues.
m_{char}^{mid}	Mid-level character memory, such as clothing or body structure.
m_{char}^{high}	High-level character memory, such as semantic identity representation.
m_{loc}	Location or background memory.
m_{prop}	Object or prop memory.
m_{style}	Global visual style memory.
m_{motion}	Motion memory extracted from temporal evidence.
m_{trans}	Transition memory extracted from boundary frames or latent states.
DAMF-U-Net notation	
$z_{i,t}$	Noisy latent video of scene s_i at diffusion timestep t.
$z_{i,0}$	Clean latent video representation of scene s_i.
t	Diffusion timestep.
$\mathbf{e}_i$	Text embedding of scene s_i.
ϵ	Gaussian noise added during forward diffusion.
$\hat{\epsilon}_{i,t}$	Predicted noise for scene s_i at timestep t.
ϵ_θ^{DAMF}	DAMF-U-Net noise prediction model.
θ	Complete model parameter set.
θ_{base}	Frozen pretrained text-to-video backbone parameters.
θ_{DAMF}	Trainable DAMF-U-Net memory-conditioning parameters.
h_l	Original U-Net feature at layer l.
$\tilde{h}_l$	Memory-guided U-Net feature at layer l.
l	U-Net layer index.
$\Phi_l^k(\cdot)$	Memory projection function for memory type k at layer l.

Continued on next page

Symbol	Description
$\lambda_l^k(t)$	Timestep-aware gate for memory type k at layer l.
ρ_l^k	Layer-wise routing weight for memory type k at layer l.
$c_{i,l,t}^k$	Effective memory coefficient for memory type k.
$W_{K,l}^k, W_{V,l}^k$	Key and value projection matrices for memory attention.
MLP_l^k	Modulation network used for appearance or style memory.
γ_l^k, β_l^k	Scale and shift parameters for FiLM-style modulation.
$\Pi(\cdot)$	Alignment projection for transition latent conditioning.
α_i^{trans}	Transition conditioning strength for scene s_i.
$\hat{\epsilon}_{i,t} = \epsilon_\theta^{DAMF}(z_{i,t}, t, \mathbf{e}_i, \mathcal{R}_i, \mathbf{g}_i, D_{j\to i})$	DAMF-U-Net noise prediction with retrieved memory and dependency strength.
$\tilde{h}_l = h_l + \sum_{k\in\mathcal{K}} g_i^k \lambda_l^k(t)\rho_l^k \Phi_l^k(m_i^k)$	Central memory-guided U-Net layer update.
Audit and repair notation	
$\mathbf{a}_i$	Audit score vector for generated scene V_i.
a_i^k	Audit score for consistency dimension k.
τ_a^k	Audit threshold for consistency dimension k.
Audit$(\cdot)$	Function that evaluates generated scene consistency.
Pass(V_i)	Binary decision indicating whether scene V_i passes audit.
Commit(μ_i^k)	Operation that commits a valid memory item to $\mathcal{B}_i$.
Reject(μ_i^k)	Operation that rejects an unreliable memory item.
Invalidate(μ_i^k)	Operation that disables outdated memory after repair.
Recovery(s_j, s_i)	Downstream recovery action after repairing source scene s_j.
$\kappa_{j\to i} < \gamma_1$	Condition for keeping the downstream scene unchanged.
$\gamma_1 \le \kappa_{j\to i} < \gamma_2$	Condition for updating downstream memory or context only.
$\kappa_{j\to i} \ge \gamma_2$	Condition for regenerating the downstream scene.
Loss notation	
$\mathcal{L}_{noise}$	Diffusion noise prediction loss.
$\mathcal{L}_{id}$	Identity consistency loss.

Continued on next page

Symbol	Description
$\mathcal{L}_{app}$	Appearance consistency loss.
$\mathcal{L}_{loc}$	Location consistency loss.
$\mathcal{L}_{prop}$	Prop consistency loss.
$\mathcal{L}_{style}$	Style consistency loss.
$\mathcal{L}_{temp}$	Temporal consistency loss.
$\mathcal{L}_{trans}$	Transition continuity loss.
$\mathcal{L}_{align}$	Text-video alignment loss.
$\mathcal{L}_{gate}$	Memory gating regularization loss.
$\mathcal{L}_{total}$	Total training objective.
λ_{id}	Weight for identity consistency loss.
λ_{app}	Weight for appearance consistency loss.
λ_{loc}	Weight for location consistency loss.
λ_{prop}	Weight for prop consistency loss.
λ_{style}	Weight for style consistency loss.
λ_{temp}	Weight for temporal consistency loss.
λ_{trans}	Weight for transition continuity loss.
λ_{align}	Weight for text-video alignment loss.
λ_{gate}	Weight for memory gating regularization loss.
Evaluation metric notation	
FVD	FrÃ©chet Video Distance.
VFID	Video FrÃ©chet Inception Distance.
CLIPScore	Text-video alignment score.
S_{id}	Identity consistency score.
S_{loc}	Location consistency score.
S_{prop}	Prop persistence score.
S_{style}	Style consistency score.
S_{motion}	Motion consistency score.
S_{trans}	Transition continuity score.

Continued on next page

Symbol	Description
S_{dep}	Dependency satisfaction score.
MCR	Memory contamination rate.
RSR	Repair success rate.
DRR	Downstream regeneration ratio.

Abstract

Long text-to-video generation remains challenging despite recent progress in diffusion-based video synthesis. Current models can generate visually plausible short clips, but long-form generation often suffers from character identity drift, appearance inconsistency, location and background changes, prop disappearance, style drift, motion inconsistency, and weak transition continuity across scenes. Existing approaches are commonly open-loop and do not provide a dependency-aware, audit-gated causal memory mechanism for deciding which visual factors should persist, which factors may change, and how failed scenes should be repaired without regenerating the entire video. To address this problem, this paper proposes DAMF-CMG, a dependency-aware closed-loop framework for long text-to-video generation. The framework combines dependency-aware scene planning, causal scene dependency modeling, constraint-aware scene packets, dependency-aware multi-level feature extraction, audit-gated causal memory, DAMF-U-Net-based generation, and closed-loop selective repair. The core generator, DAMF-U-Net, extends diffusion noise prediction beyond the noisy latent, timestep, and text embedding by incorporating current scene text, retrieved visual memory, memory conditioning strength, dependency score, timestep-aware gating, and layer-wise memory routing. The memory bank stores identity, appearance, location, prop, style, motion, and transition memories as provisional states and commits them only after audit, preventing invalid memories from contaminating future scenes. When inconsistencies occur, DAMF-CMG aims to repair failed consistency dimensions and regenerate only causally affected downstream scenes. The framework is evaluated in terms of identity consistency, location consistency, prop persistence, style consistency, transition continuity, selective repair efficiency, and dependency satisfaction. Overall, DAMF-CMG reframes long text-to-video generation as a closed-loop, memory-audited, dependency-controlled denoising process rather than independent open-loop scene generation.

Chapter 1

Introduction

1.1 Background

Text-to-video generation has become an important research direction in multimodal generative modeling because it aims to synthesize temporally coherent video sequences from natural-language descriptions. Recent advances in denoising diffusion models have significantly improved the quality and stability of generative visual synthesis by learning an iterative denoising process from noisy samples to realistic outputs [1]. Latent diffusion models further improved the computational practicality of diffusion-based generation by performing denoising in a compressed latent space while preserving strong text-guided semantic control [2]. These developments have also influenced video generation, where diffusion models are extended from static image synthesis to spatiotemporal sequence generation [3, 4]. As a result, modern text-to-video systems can generate visually plausible short clips with meaningful motion, object appearance, and prompt-conditioned visual content [5, 6]. However, generating long-form videos from text remains substantially more difficult than generating short isolated clips, because long videos require consistency across multiple semantically connected scenes rather than only local visual realism.

Long-form text-to-video generation is important for applications such as narrative storytelling, educational content creation, simulation, virtual environments, advertising, interactive media, and multimodal content production. In these scenarios, a user prompt often describes a sequence of events involving recurring characters, persistent objects, changing actions, and scene transitions. For example, a character may enter a room, pick up an object, move to another location, and later interact with the same object under a different camera view. Such generation requires the model to preserve identity, appearance,

location, object state, visual style, temporal order, and transition logic across the full narrative. Therefore, a long generated video should behave as a connected visual story rather than a simple concatenation of unrelated short video clips. This requirement makes long text-to-video generation a challenging problem involving language understanding, scene-level planning, visual memory, temporal coherence, and controllable diffusion-based synthesis.

Despite the rapid progress of text-to-video generation, long-form generation still suffers from several recurring failure modes. Character identity may drift across scenes, causing the same person to appear with different facial structure, body appearance, or clothing. Locations and backgrounds may change even when the prompt requires the same environment to continue. Important props may disappear, change shape, or reappear inconsistently. Style, lighting, and color tone may also shift between scenes, producing a visually fragmented output. Motion inconsistency and weak transition continuity can further break the perception of a coherent video sequence. These failures are especially problematic because errors in early scenes can propagate into later scenes if the generated visual state is reused without verification. Recent evaluation benchmarks also show that video generation cannot be assessed only through visual quality; it also requires evaluation of text alignment, subject consistency, motion smoothness, temporal flickering, spatial relationships, compositional understanding, and long-range coherence [7, 8].

Existing text-to-video models have made significant contributions to short video synthesis and prompt-conditioned generation. Make-A-Video and Imagen Video demonstrated that diffusion-based and cascaded generation strategies can produce high-quality text-conditioned videos [5, 6]. CogVideo and Phenaki further showed the potential of large-scale pretraining and variable-length generation from textual descriptions [9, 10]. However, many existing approaches are mainly designed around prompt-level or clip-level generation. Even when longer videos are produced, the generation process often does not explicitly represent which earlier scenes affect later scenes. As a result, the model may not know which character, object, location, style, or motion factor must persist, which factors are allowed to change, and which future scenes should be updated when an earlier scene is corrected. This limitation becomes critical in narrative generation, where scenes are not independent but are connected through semantic and visual dependencies.

Recent studies have explored LLM-guided planning, story-level generation, and consistency aware video synthesis to address parts of this problem. LLM-based video planning can decompose a long prompt into multiple scenes and generate structured information such as entities, backgrounds, layouts, and consistency groups [11]. LLM-grounded video diffusion further shows that language models can support dynamic layout planning

for downstream video generation [12]. Other consistency-oriented methods attempt to preserve subject or story-level visual content using consistent attention, reference conditioning, or memory-guided mechanisms [13]. These directions are valuable because they show that planning, layout control, and reference-based consistency can improve multi-scene generation. However, planning alone does not guarantee that the generated video actually satisfies the intended visual dependencies after generation. Similarly, consistency mechanisms that reuse references or cached visual information do not always verify whether the stored memory is correct, whether it is relevant to the current scene, or whether it should be invalidated after a failed generation. Localized refinement methods are promising for correcting text-video misalignment [14], but long-form generation additionally requires repair decisions to be linked with causal dependencies between scenes.

Therefore, a key research gap remains in long text-to-video generation. Existing methods do not fully provide a unified dependency-aware, audit-gated, memory-conditioned, and closed-loop framework that jointly models scene dependencies, verifies visual memory, injects relevant memory into the denoising process, and selectively repairs failed scenes with downstream dependency recovery. A robust long-video generation framework should explicitly determine which visual elements must persist, which elements may change, which memories are safe to reuse, and which future scenes are affected when a generated scene is repaired. Without such a mechanism, long-form generation remains largely open-loop: scenes are generated and possibly evaluated, but the system lacks a principled way to audit memory, prevent memory contamination, and regenerate only the causally affected parts of the video.

To address this gap, this paper proposes DAMF-CMG, a dependency-aware causal memory-guided framework for long text-to-video generation. The core generator of the framework is DAMF-U-Net, a Dependency-Aware Multi-Level Feature-Guided U-Net with Audit-Gated Causal Memory. The proposed framework first converts a long prompt into structured scene states rather than plain scene descriptions. Each scene state includes entities, locations, props, actions, temporal information, continuity constraints, reference requirements, and transition metadata. A repair-aware causal scene dependency graph is then constructed to identify how each scene depends on earlier scenes through identity, appearance, location, prop, style, motion, and transition relations. For each target scene, a constraint-aware scene packet defines invariant constraints, allowed changes, memory requirements, audit targets, and conditioning strengths. Dependency-aware multi-level feature extraction obtains relevant visual memories from generated scenes, while the audit-gated causal memory bank stores extracted memories first as provisional memories and commits them only after consistency verification. DAMF-U-Net then uses the current scene text, retrieved committed memory, dependency score, memory condition-

ing strength, timestep-aware gates, and layer-wise routing to guide diffusion denoising. When a scene fails audit, the framework is designed to repair the failed consistency dimensions and regenerate only causally affected downstream scenes.

The main contributions of this paper are summarized as follows:

- A transition-aware scene segmentation strategy is introduced to convert long prompts into generation-ready scene states with continuity constraints and repair metadata.
- A repair-aware causal scene dependency graph is formulated to identify typed visual dependencies and downstream repair impact across scenes.
- A dependency-aware multi-level feature extraction module is designed to extract identity, appearance, location, prop, style, motion, and transition memories according to the active dependency structure.
- An audit-gated causal memory bank is proposed to store only verified and committed visual memories, thereby reducing the risk of error propagation across long videos.
- DAMF-U-Net is introduced as a dependency-aware multi-level feature-guided U-Net that injects verified visual memory into diffusion denoising through memory conditioning strength, timestep-aware gates, and layer-wise memory routing.
- A selective scene repair and downstream regeneration mechanism is formulated for closed-loop long text-to-video generation.

The remainder of this paper is organized as follows. Section 2.2 reviews related work on diffusion models, text-to-video generation, long-form video generation, LLM-based planning, consistency-aware generation, memory-guided generation, U-Net conditioning, evaluation benchmarks, and repair methods. Section **??** presents the proposed DAMF-CMG framework and the DAMF-U-Net architecture. Section 3.2 describes the experimental setup, datasets, baselines, implementation details, and evaluation metrics. Section 3.3 reports the results and discussion. Section **??** presents ablation studies and comparison with existing methods. Finally, Section 4.2 concludes the paper with limitations and future research directions.

1.2 Motivation

“‘latex Text-to-video generation has become an important multimodal generation task because it enables natural-language descriptions to be converted into dynamic visual content. This capability is increasingly relevant for narrative storytelling, educational me-

dia, simulation, digital content creation, virtual environments, and assistive video generation. Recent text-to-video models have shown substantial progress in synthesizing visually plausible short clips from textual prompts, including approaches that transfer text-to-image generation knowledge to video generation and methods that adapt diffusion-based models for low-cost or zero-shot video synthesis. However, long-form text-to-video generation remains considerably more difficult than short-clip synthesis. A long narrative usually contains multiple scenes, recurring characters, persistent objects, changing actions, different locations, and temporal transitions. Therefore, the task is not only to generate more frames, but to preserve character identity, appearance and outfit, location and background, props, visual style, motion continuity, transition coherence, and story-level semantic consistency across dependent scenes. Recent multi-scene and story-level generation studies also indicate that maintaining consistent content across scenes is a central challenge when moving beyond single-scene video generation.

This challenge motivates the need for a generation process that is more structured than independent scene synthesis. When scenes are generated in an open-loop manner, the same character may gradually change appearance, important props may disappear, backgrounds may shift unexpectedly, and errors introduced in early scenes may influence later scenes. LLM-based planning can improve scene decomposition, entity grouping, layout construction, and narrative organization, but planning alone cannot ensure that the visual output after diffusion sampling satisfies the intended dependencies. A long-video system also requires verified visual memory, dependency-aware retrieval, memory-conditioned denoising, and a repair mechanism after failed generation. Memory is necessary because recurring characters, props, and locations require access to previous visual evidence; however, memory should not be stored blindly because failed or inconsistent scenes may contaminate future generations. This motivates audit-gated causal memory. In addition, standard diffusion U-Nets predict noise mainly from the noisy latent, timestep, and text embedding, which is insufficient when later scenes must reuse dependency-relevant visual memory. DAMF-U-Net is therefore motivated as a memory-guided denoising module that injects retrieved memory into selected U-Net layers through memory projection, memory strength, timestep gates, and layer-wise routing. Finally, because regenerating an entire long video after one failed scene is inefficient, a repair-aware dependency graph is needed to identify which downstream scenes are actually affected. These challenges motivate a dependency-aware, memory-audited, closed-loop framework that can plan scenes, preserve verified visual memory, condition denoising through DAMF-U-Net, and selectively repair failed scenes for long text-to-video generation.

1.3 Problem Statement

Long-form multi-scene text-to-video generation aims to synthesize a coherent video from a long natural-language prompt that describes a sequence of scenes, events, characters, locations, objects, actions, and transitions. This task is substantially more difficult than short-clip text-to-video generation. Short clips mainly require local prompt alignment and short-term temporal coherence, whereas long-form generation requires dependency preservation across an extended narrative. A generated long video must maintain character identity, appearance and outfit, location and background continuity, object and prop persistence, style and lighting consistency, motion continuity, transition coherence, and story-level semantic consistency across multiple related scenes.

When scenes are generated independently or only weakly connected, the output may suffer from identity drift, appearance inconsistency, object disappearance, location drift, style changes, motion discontinuity, weak scene transitions, semantic drift, and error propagation into later scenes. Existing diffusion-based text-to-video methods have made strong progress in prompt-conditioned short-video synthesis, and LLM-based planning methods can improve scene decomposition and narrative organization. However, long text-to-video generation still requires a mechanism that can identify source-to-target scene dependencies, determine what should persist and what may change, extract dependency-relevant visual memory, verify memory before reuse, condition denoising using verified memory, repair failed scenes, and regenerate only causally affected downstream scenes.

Therefore, the central problem addressed in this paper is how to generate a long video from a long prompt while preserving scene-level dependencies through verified visual memory and preventing failed or inconsistent scenes from contaminating future generation. This problem is not only a scene-planning problem. A key technical challenge is how to inject dependency-relevant and audit-verified visual memory into the diffusion denoising process itself. This motivates DAMF-CMG and its core generator DAMF-U-Net, where long text-to-video generation is treated as a dependency-controlled, memory-audited, closed-loop denoising process rather than a sequence of independent open-loop scene generations.

1.4 Objectives

The objective of this study is to develop and evaluate DAMF-CMG, a dependency-aware, audit-gated, memory-guided framework for long text-to-video generation. The specific objectives are as follows:

1. To formulate long text-to-video generation as a multi-scene dependency preserva-

tion problem, where the generated video must maintain character identity, appearance, location, prop, style, motion, and transition consistency across related scenes.

2. To design a transition-aware scene understanding module that converts a long natural-language prompt into structured scene states containing scene descriptions, entities, locations, props, actions, temporal information, continuity constraints, reference requirements, and transition metadata.

3. To construct a repair-aware causal dependency graph that identifies typed dependencies between source scenes and target scenes, including identity, appearance, location, prop, style, motion, and transition dependencies, together with downstream repair impact.

4. To develop a constraint-aware scene packet representation that specifies invariant constraints, allowed changes, memory requirements, memory-conditioning strengths, audit targets, memory update policies, and repair hints for each target scene.

5. To extract dependency-relevant multi-level visual memories and organize them through an audit-gated causal memory bank, where only verified memories are committed and unreliable or outdated memories are rejected or invalidated.

6. To retrieve only committed, valid, and dependency-compatible memories for each target scene, instead of blindly using the immediately previous scene or all stored visual memories.

7. To design DAMF-U-Net as the central memory-guided denoising module, where retrieved visual memory is injected into diffusion generation through memory projection Φ_l^k, memory strength g_i^k, timestep-aware gates $\lambda_l^k(t)$, and layer-wise routing ρ_l^k.

8. To evaluate the proposed framework using standard video-generation metrics and long-video consistency measures, including text-video alignment, identity consistency, appearance consistency, location consistency, prop persistence, style consistency, transition continuity, dependency satisfaction, memory contamination, selective repair success, and downstream regeneration efficiency.

1.5 Problem Formulation

Let the input be a long-form text prompt P, which describes a sequence of events, entities, locations, objects, actions, and temporal transitions. The objective is to generate a

long coherent video V that follows the prompt while preserving dependency-aware visual consistency across scenes. The prompt is decomposed into an ordered set of scene states:

$$S = \{s_1, s_2, \ldots, s_N\}, \tag{1.1}$$

where N is the number of scenes and s_i denotes the ith scene. Each scene state is used to generate a corresponding video segment:

$$V_i = \{x_{i,1}, x_{i,2}, \ldots, x_{i,F_i}\}, \tag{1.2}$$

where $x_{i,f}$ is the fth frame of scene s_i, and F_i is the number of frames in that scene. The final long video is obtained by concatenating all accepted scene segments:

$$V = \text{Concat}(\bar{V}_1, \bar{V}_2, \ldots, \bar{V}_N), \tag{1.3}$$

where $\bar{V}_i$ denotes the accepted or repaired version of scene V_i.

A simple scene-wise generation strategy may generate each V_i independently from its scene description. However, independent generation is insufficient for long narratives because a target scene s_i may depend on an earlier source scene s_j, where $j < i$. This dependency may arise from a repeated character, the same outfit, a persistent location, a recurring object, a continuing action, a shared style, or a transition that visually connects two scenes.

To model this relationship, the proposed framework defines a dependency score from source scene s_j to target scene s_i:

$$D_{j\to i} \in [0, 1], \quad j < i. \tag{1.4}$$

A higher value of $D_{j\to i}$ indicates that scene s_i requires stronger continuity from scene s_j. The corresponding typed dependency vector is defined as

$$\mathbf{z}_{j\to i} = [z^{id}_{j\to i}, z^{app}_{j\to i}, z^{loc}_{j\to i}, z^{prop}_{j\to i}, z^{style}_{j\to i}, z^{motion}_{j\to i}, z^{trans}_{j\to i}], \tag{1.5}$$

where each component indicates whether a specific dependency type is active. These components correspond to identity, appearance, location, prop, style, motion, and transition dependencies, respectively.

The problem can therefore be stated as follows: given a long prompt P, generate a sequence of video scenes $\{\bar{V}_i\}_{i=1}^{N}$ such that each scene is aligned with its local scene description while also satisfying the cross-scene dependency constraints induced by earlier

scenes. Conceptually, this objective can be written as

$$P \rightarrow \{\bar{V}_i\}_{i=1}^{N} \quad \text{subject to} \quad C_{text}, C_{dep}, C_{mem}, C_{repair}, \tag{1.6}$$

where C_{text} denotes scene-level text alignment, C_{dep} denotes dependency consistency, C_{mem} denotes valid memory reuse, and C_{repair} denotes selective repair consistency. These constraints are not treated only as final evaluation checks; they influence scene packet construction, memory retrieval, DAMF-U-Net denoising, audit, and downstream repair decisions.

1.6 Chapter Summary

Chapter Summary

This chapter presented the background, motivation, problem statement, and research objectives of the proposed DAMF-CMG framework for long text-to-video generation. The discussion showed that long-form multi-scene video generation is more challenging than short-clip generation because it requires persistent character identity, stable appearance, location continuity, prop preservation, style consistency, motion coherence, transition continuity, and story-level semantic consistency across dependent scenes. The chapter also highlighted the limitations of independent open-loop scene generation, where visual drift, object disappearance, weak transitions, and error propagation may occur when earlier generated scenes are not verified before influencing later scenes. The problem statement identified the central challenge as generating a coherent long video from a long natural-language prompt while preserving scene-level dependencies through verified visual memory. It emphasized that long text-to-video generation is not only a scene-planning problem, but also a memory-conditioned denoising problem in which dependency-relevant visual information must be extracted, verified, retrieved, and injected into the generation process. The research objectives then defined the main goals of the study, including transition-aware scene state construction, repair-aware causal dependency graph modeling, constraint-aware scene packet generation, audit-gated memory management, dependency-aware memory retrieval, DAMF-U-Net-based memory-guided denoising, and evaluation using both standard and dependency-specific metrics. Overall, this chapter established the foundation for the proposed dependency-controlled, memory-audited, and closed-loop approach to long text-to-video generation.

Chapter 2

Literature Review

2.1 Tools and Technologies Used

The proposed DAMF-CMG framework is implemented using a Python-based deep learning environment. The implementation uses Python [Python version], PyTorch [PV], CUDA [CUDA version], and cuDNN [cuDNN version] for GPU-accelerated model execution and training. Hugging Face Diffusers [Diffusers version] and Transformers [TV] are used where compatible with the selected text-to-video backbone and text-encoding modules. OpenCV and PIL are used for frame extraction, image processing, mask handling, and video-frame visualization. NumPy and Pandas support numerical analysis and tabular evaluation, while Matplotlib is used for plotting metric trends and ablation results. The experiments are conducted in a Conda-based environment using [Jupyter Notebook] for development and analysis.

The base video generator is a pretrained text-to-video diffusion backbone, denoted as [base T2V backbone]. This backbone provides the initial latent video denoising capability and is used as the base generator for scene-level synthesis. In the proposed setup, the pretrained backbone is kept frozen when lightweight DAMF-U-Net adapter training is used. The novelty of the framework does not lie in the pretrained backbone itself, but in the dependency-aware scene modeling, audit-gated causal memory, dependency-compatible memory retrieval, and DAMF-U-Net memory-conditioned denoising mechanism.

Scene parsing and dependency planning are supported by an LLM-based planning component. The LLM is used to decompose long prompts into structured scene states, identify entities, locations, props, actions, temporal information, continuity constraints, reference requirements, and transition metadata. It also supports dependency reasoning

and scene packet construction through structured prompts or JSON-style intermediate outputs. This component is treated as a frozen planning tool and is not claimed as the central contribution of the paper.

For visual grounding and memory extraction, the implementation uses several frozen vision models. GroundingDINO is used for phrase-grounded detection of dependency-relevant regions such as characters, props, and objects [T1]. SAM or SAM2 is used to obtain segmentation masks for detected regions [T2]. CLIP is used for text-image or text-video semantic alignment and similarity scoring [T3]. DINO or DINOv2 is used for visual feature extraction, crop-level similarity, and consistency scoring across dependent scenes [T4]. A VAE encoder/decoder is used to map videos or frames between pixel space and latent space when the selected diffusion backbone operates in latent space. These tools support detection, segmentation, feature extraction, verification, and latent representation; they are frozen support modules rather than proposed trainable contributions.

The trainable part of the proposed framework is the DAMF-U-Net memory- conditioning mechanism. The trainable components include memory projection layers Φ_l^k, memory attention adapters, FiLM-style adapters, timestep-aware gates $\lambda_l^k(t)$, layer-wise routing weights ρ_l^k, memory strength gates g_i^k, and other lightweight memory-conditioning modules. These modules learn how verified identity, appearance, location, prop, style, motion, and transition memories should influence selected denoising layers and timesteps. The base text-to-video backbone remains frozen in the lightweight setting, while the memory projection, routing, gating, and adapter modules are trained or fine-tuned.

The evaluation uses existing datasets and custom prompt benchmarks. Candidate public datasets include MSR-VTT, VATEX, and ActivityNet Captions for general short-video language evaluation and feature/audit validation. In addition, the study uses a custom long-prompt benchmark with [N] narrative prompts and a controlled dependency test set designed to evaluate identity, appearance, location, prop, style, motion, transition, and independent-scene cases. Benchmark-based evaluation may also include VBench, EvalCrafter, and T2V-CompBench where applicable [T5]. These benchmarks are used only for evaluation and comparison; the custom long-prompt benchmark is required because the proposed framework specifically targets multi-scene dependency consistency.

The evaluation tools include FVD or VFID for distribution-level video quality, CLIPScore or CLIP-based similarity for text-video alignment, and fine-grained consistency measures for identity, appearance, location/background, prop persistence, style consistency, motion consistency, and transition continuity [T6]. DAMF-CMG-specific measures include dependency satisfaction rate, memory contamination rate, repair effectiveness, and downstream regeneration ratio. No numerical results are reported in this section.

The computational setup is reported using the following reproducibility settings: GPU

model [GPU model], GPU memory [VRAM], CPU [CPU], RAM [RAM], operating system [OS], storage/cache setup [storage/cache setup], random seed [random seed], frame resolution [resolution], frames per scene [frames per scene], diffusion sampling steps [diffusion steps], scheduler [scheduler], batch size [batch size], number of generated samples per prompt [samples per prompt], maximum repair attempts [R_{max}], audit thresholds [audit thresholds], and memory confidence thresholds [memory thresholds]. Exact values are reported only after the final experimental configuration is fixed.

Table 2.1: Tools and technologies used in DAMF-CMG.

Category	**Tool/Technology**	**Role in DAMF-CMG**	**Status**
Programming environment	Python	Main implementation language.	Support
Deep learning framework	PyTorch	Model training and inference.	Support
GPU acceleration	CUDA/cuDNN	GPU-accelerated training and inference.	Support
Model framework	Diffusers / Transformers	Backbone and model integration.	Support
Base generator	Base T2V backbone	Pretrained text-to-video diffusion generator used for scene-level synthesis.	Frozen
Scene planning	LLM	Scene segmentation, dependency parsing, and scene packet construction.	Frozen
Detection	GroundingDINO	Phrase-grounded character, prop, and object detection.	Frozen
Segmentation	SAM/SAM2	Region mask extraction for dependency-relevant visual memory.	Frozen
Semantic alignment	CLIP	Text-image or text-video similarity and verification.	Frozen

Continued on next page

Category	Tool/Technology	Role in DAMF-CMG	Status
Visual features	DINO/DINOv2	Crop, background, and consistency feature extraction.	Frozen
Latent representation	VAE encoder/decoder	Pixel-latent encoding and decoding for latent diffusion backbones.	Frozen
DAMF-U-Net	Φ_l^k, adapters, gates, routing	Memory-conditioned denoising using projection, timestep-aware gating, and layer-wise routing.	Trainable
Datasets	MSR-VTT, VATEX, ActivityNet Captions	General video-language evaluation.	Evaluation
Custom benchmark	Long-prompt benchmark	Multi-scene dependency testing.	Evaluation
Metrics	FVD/VFID, CLIPScore, VBench-style metrics	Quality, alignment, and consistency evaluation.	Evaluation
Hardware	GPU/CPU/RAM/OS	Experimental computation setup.	Support

2.2 Related Work

2.2.1 Diffusion and Latent Diffusion Models

Diffusion models have become a major foundation for modern visual generation because they learn to synthesize data through an iterative denoising process. Denoising Diffusion Probabilistic Models (DDPMs) established a practical formulation in which a neural network learns to reverse a gradual noise corruption process, enabling high-quality image generation without relying on adversarial training [1]. Classifier-free guidance further improved conditional generation by allowing a diffusion model to control the strength of conditioning without requiring a separate external classifier [15]. These developments made diffusion models suitable for text-conditioned visual synthesis, where semantic guidance and generation quality must be balanced carefully.

Latent Diffusion Models (LDMs) reduced the computational burden of diffusion by moving the denoising process from pixel space to a compressed latent representation [2].

This idea became especially important for high-resolution generation because it made diffusion more practical while retaining strong text-conditioning capacity through attention-based mechanisms. The extension of diffusion models to video generation introduced additional challenges because the model must synthesize not only individual frames but also coherent temporal evolution across frames. Video Diffusion Models and latent video diffusion methods adapted diffusion-based generation to video by incorporating temporal modeling and latent-space training strategies [3, 4]. These works provide the foundation for diffusion-based video synthesis. However, they mainly define how to generate images or video clips from noise and conditioning signals; they do not directly address long multi-scene dependency modeling, audit-verified memory reuse, or selective downstream scene repair.

2.2.2 Text-to-Video Generation

Text-to-video generation extends text-conditioned image synthesis to dynamic visual sequences. Early influential methods such as Make-A-Video showed that text-to-image generation knowledge can be transferred to video synthesis by adding temporal modeling and using video data to learn motion patterns [5]. Imagen Video demonstrated high-definition text-to-video generation through cascaded diffusion modules and progressive refinement [6]. CogVideo and later CogVideoX explored large-scale text-to-video generation using transformer-based or diffusion-transformer-based designs, showing the importance of scalable architectures and large visual-language training for open-domain video generation [9, 16]. Phenaki is also important because it addressed variable-length video generation from open-domain textual descriptions, making it relevant to long-prompt and sequence-based video generation [10].

Other video generation systems, such as VideoCrafter, VideoCrafter2, and Stable Video Diffusion, further improved video synthesis through latent video modeling, image-to-video generation, and stronger motion representation [17, 18]. These methods have advanced the quality, diversity, and controllability of generated videos. Nevertheless, many text-to-video models remain strongest at short-clip or prompt-level generation. When a long narrative requires the same character, outfit, background, object, style, motion state, or transition to persist across multiple scenes, prompt conditioning alone may not provide sufficient control. This limitation motivates approaches that explicitly model what information should be preserved, how it should be stored, and how it should guide later denoising steps.

2.2.3 Long-Form and Multi-Scene Video Generation

Long-form and multi-scene video generation focuses on producing videos that remain coherent over several events or shots. Compared with short-clip synthesis, this setting requires a stronger representation of narrative order, scene transitions, entity persistence, and visual continuity. VideoDirectorGPT uses large language models to support multi-scene video generation by planning scenes, entities, backgrounds, layouts, and consistency information [11]. LLM-grounded Video Diffusion Models use language models to generate dynamic layout conditions that guide video diffusion, showing that structured language-based planning can improve controllability [12]. These works demonstrate that language models can be useful for decomposing complex prompts and organizing visual generation into more manageable sub-problems.

Recent story-level and multi-shot methods have also explored consistency across longer sequences. StoryDiffusion improves long-range content consistency by using mechanisms that encourage subjects to remain stable across generated images or videos [13]. Video Storyboarding and related multi-shot generation methods focus on maintaining character consistency across multiple shots, which is especially relevant for narrative video generation [19]. ShotAdapter and Corgi are also related because they investigate multi-shot generation and memory-guided consistency for complex video sequences [20, 21]. MovieBench provides a benchmark direction for movie-level generation and highlights the need to evaluate long-range story structure and character consistency [22]. These studies contribute important components for long-video generation, including planning, story structure, consistency, and memory. However, long-form generation still requires a unified mechanism that connects scene dependency, memory validity, U-Net conditioning, and downstream repair decisions.

2.2.4 LLM-Based Video Planning and Scene Decomposition

Large language models have increasingly been used as planning components in generative video pipelines. Their role is useful because long prompts often contain multiple events, characters, references, temporal relations, and implied dependencies. LLM-based planning can decompose a prompt into scene descriptions, identify entities, propose layouts, and organize generation steps before visual synthesis begins [11, 12]. In multi-scene generation, this planning stage can reduce ambiguity and provide more structured control than using the entire long prompt directly as a single text condition.

However, planning and generation are different problems. A language model may produce a reasonable sequence of scenes, but the generated video may still fail to preserve the intended visual details after diffusion sampling. For example, a planned char-

acter may drift in appearance, a planned object may disappear, or a planned location may change unexpectedly. Therefore, scene planning alone is not enough for verified long-video consistency. DAMF-CMG is positioned differently: it uses structured planning not only to describe scenes but also to support generation-ready scene packets, dependency detection, memory requirements, audit targets, and repair impact. This distinction is important because the proposed framework treats planning outputs as control information for memory-guided denoising and closed-loop repair, rather than as only textual scene descriptions.

2.2.5 Visual Consistency in Generated Videos

Visual consistency is a central challenge in long text-to-video generation. Subject-driven and reference-guided methods have made progress in preserving identity and appearance across generated visual content. DreamVideo, DreamVideo-2, VideoBooth, and MotionBooth investigate different forms of subject customization, reference conditioning, or motion-aware generation [23–26]. These works are relevant because they show that subject identity, appearance, and motion can be improved when additional reference information is available. StoryDiffusion and Video Storyboarding also contribute to this direction by focusing on long-range subject consistency and multi-shot character preservation [13, 19].

However, long-form video consistency is broader than subject identity alone. A generated video may need to preserve not only who appears in the scene, but also what the person wears, where the scene takes place, which object is being carried, what visual style continues, how motion evolves, and how one scene transitions into the next. Some methods improve identity or reference-based appearance, but they do not always model multiple dependency types together. In contrast, DAMF-CMG is designed around several dependency dimensions, including identity, appearance, location, prop, style, motion, and transition. The goal is not to replace subject-consistency methods, but to place visual consistency inside a causal memory-guided framework where each consistency requirement can be detected, stored, audited, retrieved, and repaired.

2.2.6 Memory and Retrieval-Based Generation

Memory and retrieval mechanisms have been studied as ways to provide additional visual or semantic context during generation. Retrieval-augmented generation methods use external examples or retrieved representations to support synthesis, and retrieval-augmented diffusion models show that stored visual information can help guide generative models [27]. Re-Imagen demonstrates the usefulness of retrieved image-text information for im-

proving text-to-image generation [28]. Reference-based methods such as IP-Adapter also show that visual prompts can be injected into diffusion models through adapter mechanisms while preserving text-conditioning capacity [29]. In long-video settings, cached or scene-level memory can help preserve information across multiple shots or scenes [21].

Feedback and memory have also been explored in broader reasoning and refinement systems. Self-Refine uses iterative feedback and refinement, while Reflexion studies memory-based self-improvement in agentic settings [30, 31]. These works are not text-to-video generation methods in the strict sense, but they are useful background for understanding why feedback and memory can support iterative improvement. The memory mechanism proposed in DAMF-CMG differs from generic retrieval or feedback memory. It is intended to be typed, causal, audit-gated, dependency-aware, and repair-sensitive. Memories are not reused simply because they exist; they must be relevant to the current dependency relation, pass audit before commitment, and be invalidated or updated when a repaired scene changes the causal visual state.

2.2.7 U-Net Conditioning and Adapter-Based Diffusion Control

Controllable diffusion generation has benefited from adapter-based and conditional-control mechanisms. ControlNet introduced a way to add trainable control branches to diffusion models while preserving the knowledge of a pretrained backbone [32]. T2I-Adapter similarly explored lightweight adapters for adding external control signals to text-to-image diffusion [33]. IP-Adapter proposed a decoupled image-prompt adapter, showing that visual reference information can be introduced without fully replacing text conditioning [29]. AnimateDiff added motion modules to personalized text-to-image diffusion models, enabling the generation of animated content while reusing strong image-generation priors [34]. These works show that diffusion backbones can be guided by additional signals such as spatial maps, image prompts, motion modules, or adapter features.

DAMF-U-Net is related to this family of conditioning methods because it also modifies how additional information influences denoising. However, its conditioning objective is different. Instead of using a generic external condition, DAMF-U-Net is designed to inject dependency-relevant visual memory selected from an audit-gated causal memory bank. The influence of memory is controlled by dependency strength, memory confidence, timestep-aware gates, and layer-wise routing. This allows different memory types to affect different parts of the denoising process. For example, identity, appearance, location, prop, style, motion, and transition memories do not need to enter the U-Net in the same way or with the same strength. The proposed direction therefore extends adapter-style control toward dependency-aware memory-guided denoising for long multi-scene generation.

2.2.8 Video Generation Evaluation and Repair

Evaluation is essential for long text-to-video generation because visual quality alone does not capture all relevant failure modes. FrÃ©chet Video Distance (FVD) has been widely used to measure video generation quality by comparing distributions of generated and real videos in a learned feature space [35]. CLIP and CLIPScore-style measures are often used to estimate text-image or text-video alignment through visual-language embeddings [36, 37]. More recent benchmarks provide more fine-grained evaluation. VBench evaluates video generation across several dimensions such as subject consistency, background consistency, motion smoothness, imaging quality, and semantic alignment [7]. EvalCrafter also studies multi-aspect evaluation for generated videos [38]. T2V-CompBench focuses on compositional text-to-video evaluation, including object, attribute, motion, and spatial relationship understanding [8]. MovieBench extends the evaluation discussion toward long-video and movie-level generation, where narrative structure and character consistency become important [22].

Repair and correction methods are also relevant because generation failures are common in complex text-to-video tasks. VideoRepair studies misalignment detection and localized refinement for text-to-video generation, showing that generated videos can be corrected after identifying specific text-video mismatches [14]. Self-correcting diffusion and feedback-based refinement methods provide additional background for closed-loop correction [30, 31, 39]. However, long multi-scene generation requires repair to be connected with dependency structure. If an earlier scene is repaired, some later scenes may remain valid, while others may need memory updates or regeneration. Therefore, DAMF-CMG considers evaluation and repair not only as final-stage quality checks but also as part of a closed-loop process involving audit-gated memory, dependency satisfaction, transition continuity, repair success, memory contamination control, and downstream regeneration ratio.

2.2.9 Summary of Research Gap

Existing studies address important parts of the long text-to-video generation problem. Diffusion and latent diffusion models provide the denoising foundation for high-quality visual synthesis; text-to-video methods extend this capability to prompt-conditioned video generation; LLM-guided methods improve scene planning, layout control, and story decomposition; subject-consistency and reference-guided methods improve identity or appearance preservation; memory and retrieval-based methods introduce external or cached visual support; adapter-based methods show how additional conditions can guide diffusion backbones; and repair-oriented methods demonstrate that localized correction is

possible. However, these directions have not yet fully integrated dependency-aware scene modeling, audit-gated causal memory, dependency-specific visual memory retrieval, DAMF -U-Net-style timestep-aware and layer-aware memory-conditioned denoising, and selective downstream regeneration into a single closed-loop framework. This gap motivates DAMF-CMG, which treats long text-to-video generation as a dependency-controlled and memory-audited generation process rather than a sequence of independent open-loop scene generations.

2.3 Chapter Summary

In this chapter, the main ideas and tools used in text-to-video generation were reviewed, with emphasis on the models and methods most closely related to long multi-scene generation. The literature shows that most progress has been achieved in short video clip generation. Models such as Make-A-Video, Phenaki, CogVideo, CogVideoX, VideoCrafter, and latent-diffusion-based video models have improved visual quality and prompt- conditioned generation, but long-form generation still remains difficult when a prompt describes multiple dependent scenes.

The review also shows that several tools and model components repeatedly appear in this research area, including diffusion models, latent diffusion, CLIP-based alignment, video VAEs, adapter-based conditioning, FiLM-style modulation, and video processing tools. These technologies provide the foundation for modern text-to-video systems. However, the literature also indicates that long-video generation is not only a matter of increasing model size or generating more frames. It requires better scene structuring, explicit dependency modeling, verified visual memory, and mechanisms for preserving characters, locations, props, style, motion, and transitions across scenes.

Overall, the related work identifies a clear research gap: existing methods address diffusion-based synthesis, text-to-video generation, LLM-based planning, reference- guided consistency, memory-assisted generation, adapter-based conditioning, and repair separately, but they do not fully integrate these components into one dependency-aware, audit-gated, memory-conditioned, and closed-loop framework. This gap motivates the proposed DAMF-CMG framework and DAMF-U-Net architecture.

Chapter 3

Proposed Works

3.1 Proposed Works

3.1.1 Overview of the Proposed DAMF-CMG Framework

The proposed DAMF-CMG framework is designed for long multi-scene text-to-video generation, where the objective is not only to synthesize visually plausible video segments, but also to preserve dependency-aware continuity across an extended narrative. In long-form prompts, later scenes often depend on earlier scenes through recurring characters, persistent appearance attributes, repeated locations, shared objects, visual style, motion flow, and scene transitions. If each scene is generated independently, these dependencies may be lost, resulting in identity drift, object disappearance, inconsistent backgrounds, weak transitions, and propagation of earlier generation errors. DAMF-CMG therefore formulates long text-to-video generation as a closed-loop, memory-audited, dependency-controlled denoising process.

Given a long text prompt P, the framework first decomposes the prompt into an ordered sequence of structured scene states rather than plain text fragments. Each scene state contains the scene description, entities, location, objects, action, temporal information, continuity constraints, reference requirements, and transition metadata. A repair-aware causal scene dependency graph is then constructed to identify how a target scene s_i depends on one or more earlier source scenes s_j, where $j < i$. This graph stores typed visual dependencies such as identity, appearance, location, prop, style, motion, and transition dependencies. It also estimates downstream repair impact, which is later used to decide whether repairing an earlier scene should keep, update, or regenerate later depen-

dent scenes.

For each target scene, DAMF-CMG constructs a constraint-aware scene packet P_i. The scene packet acts as the interface between language-level planning and DAMF-U-Net generation. It specifies what must remain invariant, what is allowed to change, which memory types are required, how strongly each memory type should influence denoising, which audit checks are required after generation, and which repair actions should be used if the generated scene fails. After a scene video V_i is generated, dependency-aware multi-level feature extraction obtains visual memories from the generated video. These memories include identity, appearance, location, prop, style, motion, and transition information. The extraction process is dependency-aware: only memories relevant to future dependent scenes are selected, reducing the risk of irrelevant or excessive conditioning.

A key component of DAMF-CMG is the audit-gated causal memory bank. Extracted memories are not immediately reused. They are first stored as provisional memories and are committed only if the generated scene passes the required audit checks. Memories that fail audit are rejected, and previously committed memories may be invalidated if a repaired scene changes the corresponding visual state. This mechanism reduces memory contamination and prevents incorrect visual information from influencing later scenes. For each target scene, only committed, valid, and dependency-compatible memories are retrieved from the memory bank.

The core generator of the proposed framework is DAMF-U-Net, a -Aware Multi-Level Feature-Guided U-Net. Unlike a standard text-to-video diffusion U-Net that predicts noise mainly from the noisy latent, timestep, and text embedding, DAMF-U-Net predicts noise using the current scene text, retrieved visual memory, memory conditioning strength, dependency score, timestep-aware gates, and layer-wise memory routing. Thus, the denoising process is controlled by both the semantic requirement of the current scene and the causal visual dependencies inherited from earlier scenes.

The complete closed-loop architecture of the proposed DAMF-CMG framework is illustrated in Fig. 3.1.8.

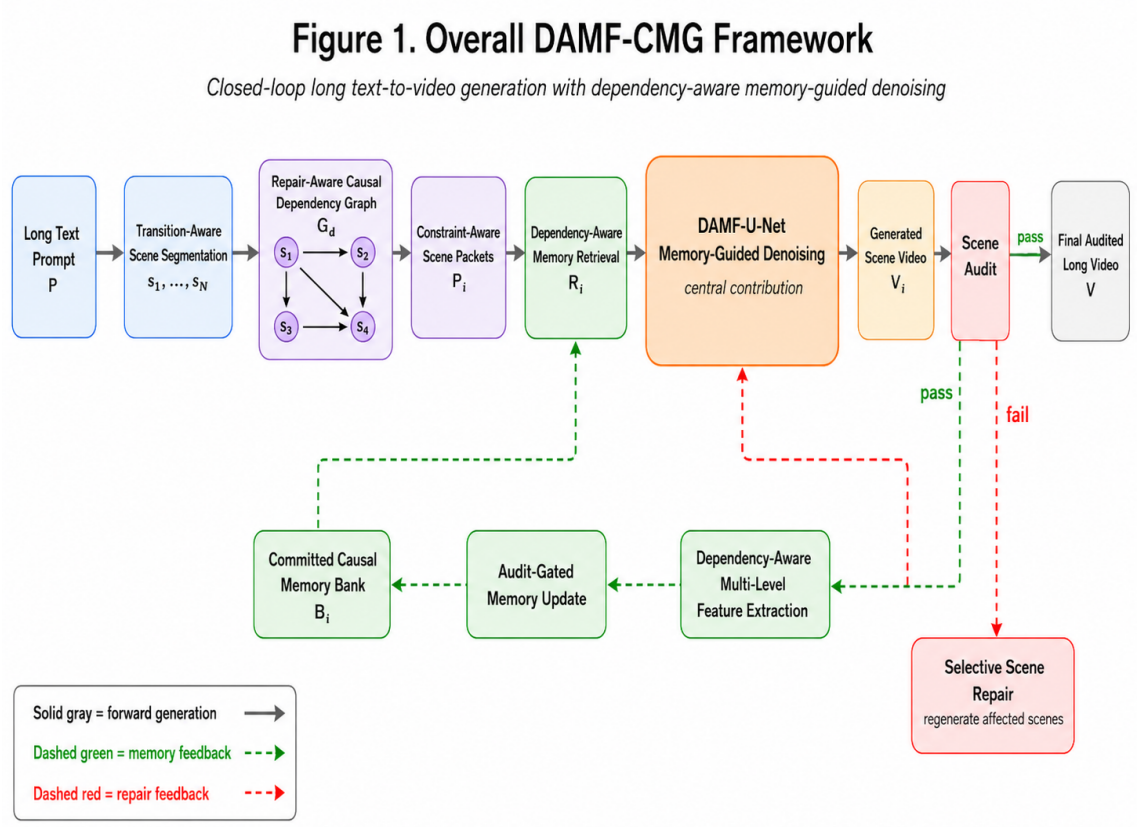


Figure 3.1: Overall DAMF-CMG framework. The proposed framework converts a long text prompt into transition-aware scene states, constructs a repair-aware causal dependency graph, and builds constraint-aware scene packets for each scene. Dependency-relevant committed memories are retrieved from the audit-gated causal memory bank and injected into DAMF-U-Net for memory-guided denoising. Generated scenes are audited before memory commitment. Passed scenes update the committed memory bank and contribute to the final long video, whereas failed scenes trigger selective repair and regeneration of only causally affected downstream scenes.

The overall closed-loop flow of the proposed framework is summarized as

$$P \rightarrow S \rightarrow G_d \rightarrow P_i \rightarrow \mathcal{R}_i \rightarrow \epsilon_\theta^{DAMF} \rightarrow V_i \rightarrow \text{Audit}(V_i) \rightarrow \mathcal{B}_i \rightarrow \text{Repair/Con}\,. \quad (3.1)$$

Here, P denotes the long prompt, S is the ordered set of structured scene states, G_d is the repair-aware causal dependency graph, P_i is the constraint-aware scene packet, $\mathcal{R}_i$ is the retrieved memory set, ϵ_θ^{DAMF} is the DAMF-U-Net denoiser, V_i is the generated scene video, and $\mathcal{B}_i$ is the updated audit-gated memory bank. “‘

3.1.2 Problem Formulation

Let the input be a long-form text prompt P, which describes a sequence of events, entities, locations, objects, actions, and temporal transitions. The objective is to generate a long coherent video V that follows the prompt while preserving dependency-aware visual consistency across scenes. The prompt is decomposed into an ordered set of scene states:

$$S = \{s_1, s_2, \ldots, s_N\}, \tag{3.2}$$

where N is the number of scenes and s_i denotes the ith scene. Each scene state is used to generate a corresponding video segment:

$$V_i = \{x_{i,1}, x_{i,2}, \ldots, x_{i,F_i}\}, \tag{3.3}$$

where $x_{i,f}$ is the fth frame of scene s_i, and F_i is the number of frames in that scene. The final long video is obtained by concatenating all accepted scene segments:

$$V = \text{Concat}(\bar{V}_1, \bar{V}_2, \ldots, \bar{V}_N), \tag{3.4}$$

where $\bar{V}_i$ denotes the accepted or repaired version of scene V_i.

A simple scene-wise generation strategy may generate each V_i independently from its scene description. However, independent generation is insufficient for long narratives because a target scene s_i may depend on an earlier source scene s_j, where $j < i$. This dependency may arise from a repeated character, the same outfit, a persistent location, a recurring object, a continuing action, a shared style, or a transition that visually connects two scenes.

To model this relationship, the proposed framework defines a dependency score from source scene s_j to target scene s_i:

$$D_{j\to i} \in [0, 1], \quad j < i. \tag{3.5}$$

A higher value of $D_{j\to i}$ indicates that scene s_i requires stronger continuity from scene s_j. The corresponding typed dependency vector is defined as

$$\mathbf{z}_{j\to i} = [z^{id}_{j\to i}, z^{app}_{j\to i}, z^{loc}_{j\to i}, z^{prop}_{j\to i}, z^{style}_{j\to i}, z^{motion}_{j\to i}, z^{trans}_{j\to i}], \tag{3.6}$$

where each component indicates whether a specific dependency type is active. These components correspond to identity, appearance, location, prop, style, motion, and transition dependencies, respectively.

The problem can therefore be stated as follows: given a long prompt P, generate a

sequence of video scenes $\{\bar{V}_i\}_{i=1}^{N}$ such that each scene is aligned with its local scene description while also satisfying the cross-scene dependency constraints induced by earlier scenes. Conceptually, this objective can be written as

$$P \rightarrow \{\bar{V}_i\}_{i=1}^{N} \quad \text{subject to} \quad C_{text}, C_{dep}, C_{mem}, C_{repair}, \tag{3.7}$$

where C_{text} denotes scene-level text alignment, C_{dep} denotes dependency consistency, C_{mem} denotes valid memory reuse, and C_{repair} denotes selective repair consistency. These constraints are not treated only as final evaluation checks; they influence scene packet construction, memory retrieval, DAMF-U-Net denoising, audit, and downstream repair decisions.

3.1.3 Transition-Aware Scene State Construction

The first stage of DAMF-CMG converts the input prompt into structured scene states. This step is necessary because a long prompt usually contains multiple events, implicit references, recurring entities, temporal relations, and continuity requirements. If the prompt is divided only into plain textual segments, the downstream generator may not know which entities must persist, which objects should reappear, which location should continue, or whether the next scene should begin from the previous scene ending.

Each scene state is defined as

$$s_i = \{d_i, E_i, L_i, O_i, A_i, T_i, C_i, R_i, \Delta_i\}. \tag{3.8}$$

Here, d_i is the natural-language scene description, E_i denotes entities or characters, L_i denotes location or background, O_i denotes objects or props, A_i denotes the main action, T_i denotes temporal information, C_i denotes continuity constraints, R_i denotes reference requirements, and Δ_i denotes the transition state from the previous scene.

The transition state is defined as

$$\Delta_i = \begin{cases} \varnothing, & i = 1, \\ \phi(s_{i-1}, s_i), & i > 1, \end{cases} \tag{3.9}$$

where $\phi(\cdot)$ is a transition extraction function that analyzes the relationship between the previous scene and the current scene. The transition state may represent an independent cut, continuous action, location change, time jump, object reappearance, or previous-frame continuity. This distinction is important because a hard cut may allow independent generation, while continuous action or object reappearance may require strong memory or transition conditioning.

The structured scene state also supports repair. Since every scene contains continuity constraints and reference requirements, the system can later identify which scenes may be affected when an earlier scene is modified. Thus, segmentation is not treated as ordinary scene splitting; it becomes the first stage of the closed-loop dependency and repair mechanism.

Algorithm 1. Transition-aware scene state construction
Input: Long prompt P, scene parser F_{seg}
Output: Structured scene states S

1. Parse the long prompt into ordered scene descriptions.
2. For each scene description d_i, extract E_i, L_i, O_i, A_i, and T_i.
3. Estimate continuity constraints C_i, including identity, appearance, location, prop, style, motion, and transition constraints.
4. Estimate reference requirements R_i, including identity, location, prop, style, motion, and previous-frame references.
5. Set $\Delta_1 = \varnothing$ for the first scene.
6. For each $i > 1$, compute $\Delta_i = \phi(s_{i-1}, s_i)$.
7. Attach repair-relevant metadata.
8. Return $S = \{s_1, s_2, \ldots, s_N\}$.

3.1.4 Repair-Aware Causal Scene Dependency Detection

After constructing structured scene states, DAMF-CMG builds a repair-aware causal dependency graph. The purpose of this graph is to identify how earlier scenes influence later scenes and how repairs should propagate across the generated video. Let

$$G_d = (V_d, E_d) \tag{3.10}$$

denote the dependency graph, where $V_d = \{s_1, s_2, \ldots, s_N\}$ is the set of scene nodes and E_d is the set of directed dependency edges. A directed edge from s_j to s_i is added when the dependency score exceeds a threshold:

$$E_d = \{e_{j\to i} \mid j < i,\ D_{j\to i} \geq \theta_d\}. \tag{3.11}$$

Each edge is represented as

$$e_{j\to i} = (s_j, s_i, \mathbf{z}_{j\to i}, \rho_{j\to i}, \kappa_{j\to i}), \tag{3.12}$$

where $\mathbf{z}_{j\to i}$ is the typed dependency vector, $\rho_{j\to i}$ is the dependency strength, and $\kappa_{j\to i}$ is the downstream repair impact.

The dependency score is computed using semantic and structural overlap between two scene states:

$$\begin{aligned} D_{j\to i} =& \lambda_E J(E_j, E_i) + \lambda_L J(L_j, L_i) + \lambda_O J(O_j, O_i) \\ &+ \lambda_A \,\mathrm{Sim}(A_j, A_i) + \lambda_T \,\mathrm{Sim}(T_j, T_i) + \lambda_S \,\mathrm{Sim}(d_j, d_i) + \lambda_\Delta I(\Delta_i). \end{aligned} \tag{3.13}$$

Here, $J(\cdot,\cdot)$ denotes set-level overlap, $\mathrm{Sim}(\cdot,\cdot)$ denotes semantic similarity, and $I(\Delta_i)$ is a transition-continuity indicator. For entity, location, and object overlap, Jaccard similarity is used:

$$J(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}. \tag{3.14}$$

The dependency strength is classified as

$$\rho_{j\to i} = \begin{cases} 0, & D_{j\to i} < \theta_1, \\ 1, & \theta_1 \le D_{j\to i} < \theta_2, \\ 2, & D_{j\to i} \ge \theta_2. \end{cases} \tag{3.15}$$

Here, $\rho_{j\to i} = 0$ indicates an independent scene pair, $\rho_{j\to i} = 1$ indicates a soft dependency, and $\rho_{j\to i} = 2$ indicates a hard dependency. A soft dependency may require weak memory or symbolic continuity support, while a hard dependency requires stronger visual memory conditioning.

The repair impact from source scene s_j to downstream target scene s_i is defined as

$$\kappa_{j\to i} = \eta_{id} z^{id}_{j\to i} + \eta_{loc} z^{loc}_{j\to i} + \eta_{prop} z^{prop}_{j\to i} + \eta_{motion} z^{motion}_{j\to i} + \eta_{trans} z^{trans}_{j\to i}. \tag{3.16}$$

The corresponding downstream recovery action is

$$\mathrm{Recovery}(s_j, s_i) = \begin{cases} \text{keep}, & \kappa_{j\to i} < \gamma_1, \\ \text{update_context}, & \gamma_1 \le \kappa_{j\to i} < \gamma_2, \\ \text{regenerate}, & \kappa_{j\to i} \ge \gamma_2. \end{cases} \tag{3.17}$$

This rule prevents unnecessary full-video regeneration. If the repair impact is low, the downstream scene is kept unchanged. If the impact is moderate, only the memory context

is updated. If the impact is high, the downstream scene is regenerated.

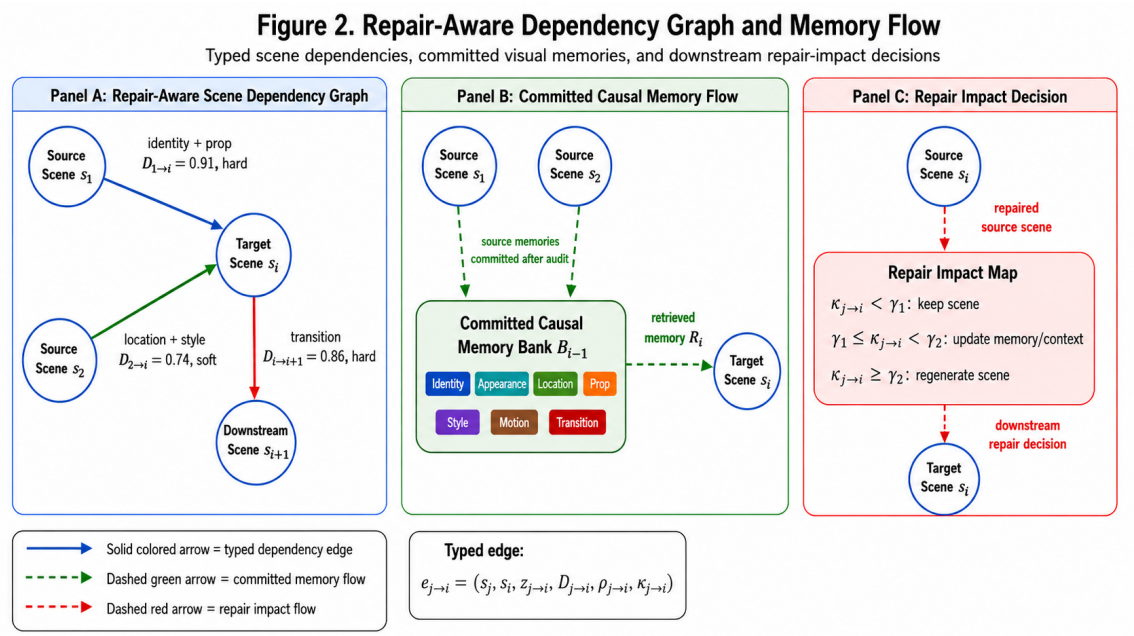


Figure 3.2: Repair-aware dependency graph and memory flow. This figure should show source scenes, target scenes, typed dependency edges, dependency scores, memory types, committed memory, and repair impact.

3.1.5 Constraint-Aware Scene Packet Builder

The scene packet builder converts each target scene s_i into a control representation for DAMF-U-Net generation. It translates the scene state and dependency graph into explicit instructions: what should be preserved, what may change, which memories are required, how strongly each memory should influence denoising, which audit checks must be applied, and which repair actions may be used if generation fails.

Let $\mathcal{E}_i$ denote the set of incoming dependency edges for scene s_i:

$$\mathcal{E}_i = \{e_{j\to i} \mid j < i,\ e_{j\to i} \in E_d\}. \tag{3.18}$$

The scene packet is constructed as

$$P_i = \Psi(s_i, \mathcal{E}_i, \mathcal{B}_{i-1}), \tag{3.19}$$

where $\Psi(\cdot)$ is the packet construction function and $\mathcal{B}_{i-1}$ is the committed memory bank available before generating scene s_i.

The complete scene packet is defined as

$$P_i = \{\Omega_i, \mathcal{I}_i, \mathcal{A}_i^{chg}, \mathbf{q}_i, \mathbf{r}_i, \mathbf{g}_i, \mathcal{G}_i, \mathcal{Q}_i, \mathcal{U}_i, \mathcal{H}_i\}. \tag{3.20}$$

Here, Ω_i is the observation state, $\mathcal{I}_i$ denotes invariant constraints, $\mathcal{A}_i^{chg}$ denotes allowed-change constraints, $\mathbf{q}_i$ is the continuity priority vector, $\mathbf{r}_i$ is the memory requirement vector, $\mathbf{g}_i$ is the memory conditioning strength vector, $\mathcal{G}_i$ is the generation contract, $\mathcal{Q}_i$ denotes audit targets, $\mathcal{U}_i$ is the memory update policy, and $\mathcal{H}_i$ contains repair hints.

For each memory type $k \in \mathcal{K}$, where

$$\mathcal{K} = \{\text{id}, \text{app}, \text{loc}, \text{prop}, \text{style}, \text{motion}, \text{trans}\}, \tag{3.21}$$

the scene packet assigns a continuity priority value q_i^k. This value estimates how important memory type k is for generating scene s_i:

$$q_i^k = q_0^k + \alpha_1 \mathbb{1}(I_i^k) + \alpha_2 z_{j\to i}^k + \alpha_3 D_{j\to i} - \alpha_4 \mathbb{1}(A_i^{chg,k}). \tag{3.22}$$

Here, q_0^k is the base priority of memory type k, $\mathbb{1}(I_i^k)$ indicates whether factor k must remain invariant, $z_{j\to i}^k$ is the dependency-type indicator, $D_{j\to i}$ is the dependency score, and $\mathbb{1}(A_i^{chg,k})$ indicates whether factor k is allowed or required to change. The subtraction term prevents memory from dominating when the current scene intentionally changes that factor.

The memory requirement vector is obtained as

$$r_i^k = \begin{cases} 1, & q_i^k \geq \theta_r, \\ 0, & q_i^k < \theta_r. \end{cases} \tag{3.23}$$

If $r_i^k = 1$, memory type k is required for the current scene. The preliminary memory conditioning strength is computed as

$$g_i^k = q_i^k \cdot \bar{\rho}_{j\to i} \cdot \text{Conf}(\mu_j^k) \cdot (1 - \chi_i^k), \tag{3.24}$$

where $\bar{\rho}_{j\to i}$ is the normalized dependency strength, $\text{Conf}(\mu_j^k)$ is the confidence of retrieved memory, and χ_i^k is a conflict indicator. If the current scene requires a change in factor k, then χ_i^k increases and the corresponding memory influence is reduced.

The scene packet therefore serves as the bridge between dependency modeling and memory-guided denoising. It tells the feature extractor which visual factors should be extracted, tells the retrieval module which committed memories are required, tells DAMF-U-Net how strongly each memory type should influence denoising, and tells the audit

module what must be verified after generation.

3.1.6 Dependency-Aware Multi-Level Feature Extraction

After a scene video is generated, DAMF-CMG extracts structured visual memory from it. Unlike a simple previous-frame reference strategy, the proposed feature extraction module does not reuse the whole generated scene or every frame indiscriminately. Instead, it uses the scene packet, dependency graph, and memory requirement vector to decide which visual factors should be extracted.

The dependency-aware extraction function is defined as

$$\mathcal{M}_i = \mathcal{F}_{extract}(V_i, P_i, \mathcal{E}_i, \mathbf{r}_i), \tag{3.25}$$

where $\mathcal{M}_i$ is the extracted memory set, V_i is the generated scene video, P_i is the scene packet, $\mathcal{E}_i$ is the incoming dependency edge set, and $\mathbf{r}_i$ specifies which memory types are required.

The memory set for scene s_i is represented as

$$\mathcal{M}_i = \{m_i^{id}, m_i^{app}, m_i^{loc}, m_i^{prop}, m_i^{style}, m_i^{motion}, m_i^{trans}\}. \tag{3.26}$$

Each memory type is extracted only when it is required by the scene packet or by future dependency requirements. For each required memory type k, a phrase query q_i^k is obtained from the scene packet. For character memory, this phrase may contain the character name, appearance, clothing, or role. For location memory, it may describe the background or environment. For prop memory, it may describe a recurring object. Phrase-grounded extraction prevents the system from selecting the wrong character or object when multiple visual entities are present.

For candidate frame $x_{i,f}$, a memory-specific frame score is computed as

$$Score_k(x_{i,f}) = \omega_1^k Q_{det} + \omega_2^k Q_{vis} + \omega_3^k Q_{cen} + \omega_4^k Q_{sharp} + \omega_5^k Q_{sem} - \omega_6^k Q_{occ}. \tag{3.27}$$

Here, Q_{det} is detection confidence, Q_{vis} is visibility quality, Q_{cen} measures spatial centrality, Q_{sharp} measures sharpness, Q_{sem} measures semantic alignment with the phrase query, and Q_{occ} is an occlusion penalty. The best reference frame for memory type k is selected as

$$f_k^* = \arg\max_{1 \le f \le F_i} Score_k(x_{i,f}). \tag{3.28}$$

For character and prop memory, phrase-grounded detection and segmentation are used

to isolate the relevant region:

$$b_i^k = \text{Detector}(x_{i,f_k^*}, q_i^k), \quad M_i^k = \text{Segment}(x_{i,f_k^*}, b_i^k), \quad \tilde{x}_i^k = x_{i,f_k^*} \odot M_i^k. \tag{3.29}$$

For location memory, foreground regions are suppressed so that background information is emphasized. For style memory, the full representative frame is used because style is a global visual property. For motion memory, a short temporal clip is used instead of a single frame. For transition memory, the last frame or boundary latent state is retained when the next scene requires visual boundary continuity.

Each selected memory is decomposed into low-level, mid-level, and high-level representations:

$$m_i^k = \{m_{i,low}^k, m_{i,mid}^k, m_{i,high}^k\}. \tag{3.30}$$

Low-level features capture local appearance details such as color, texture, edges, boundaries, hair outline, and clothing patterns. Mid-level features capture part-level and structural information such as body configuration, object structure, clothing layout, and spatial organization. High-level features encode semantic identity or conceptual representation. Motion memory is extracted from a temporal window, while transition memory is extracted from boundary frames or latent states.

Each extracted memory candidate receives a confidence score:

$$\text{Conf}(m_i^k) = \nu_1^k Q_{det} + \nu_2^k Q_{sem} + \nu_3^k Q_{vis} + \nu_4^k Q_{sharp} - \nu_5^k Q_{occ}. \tag{3.31}$$

The memory candidate is retained only if

$$\text{Conf}(m_i^k) \geq \tau_k, \tag{3.32}$$

where τ_k is the confidence threshold for memory type k. If the confidence score is below the threshold, the candidate is rejected or an alternative frame is selected.

3.1.7 Audit-Gated Causal Memory Bank

The extracted memories are not immediately reused for future scene generation. A generated scene may contain an incorrect face, missing object, inconsistent background, weak style match, or poor transition continuity. If such information is directly stored and reused, the error may propagate across later scenes. To reduce this risk, DAMF-CMG introduces an audit-gated causal memory bank.

Let $\mathcal{B}_{i-1}$ denote the committed memory bank before scene s_i, and let $\mathcal{M}_i$ denote mem-

ory candidates extracted from scene V_i. The memory bank is updated as

$$\mathcal{B}_i = \text{Update}(\mathcal{B}_{i-1}, \mathcal{M}_i, \mathbf{a}_i, G_d, P_i), \tag{3.33}$$

where $\mathbf{a}_i$ is the scene audit vector, G_d is the dependency graph, and P_i is the scene packet.

Each memory item of type k extracted from scene s_i is represented as

$$\mu_i^k = \{id_i^k, source_i, type^k, f_i^k, meta_i^k, state_i^k, audit_i^k, valid_i^k, conf_i^k, links_i^k, ver_i^k\}. \tag{3.34}$$

Here, id_i^k is the memory identifier, $source_i$ is the source scene, $type^k$ is the memory type, f_i^k is the feature tensor, $meta_i^k$ stores metadata such as phrase, frame index, bounding box, and mask, $state_i^k$ is the memory state, $valid_i^k$ indicates whether the memory can be retrieved, $conf_i^k$ is the confidence score, $links_i^k$ stores causal dependency links, and ver_i^k denotes the memory version.

Each memory follows the state lifecycle

$$state(\mu_i^k) \in \{\text{provisional}, \text{committed}, \text{rejected}, \text{invalidated}\}. \tag{3.35}$$

A newly extracted memory is first stored as provisional and is not eligible for retrieval. It becomes usable only if it passes the required audit dimensions and confidence threshold. Let Q_i^k denote the audit dimensions required for committing memory type k. The audit pass condition is

$$\text{AuditPass}(\mu_i^k) = \prod_{q \in Q_i^k} \mathbb{1}(a_i^q \geq \theta_i^q). \tag{3.36}$$

The memory is committed only if

$$\text{Commit}(\mu_i^k) = \begin{cases} 1, & \text{AuditPass}(\mu_i^k) = 1 \wedge \text{Conf}(\mu_i^k) \geq \tau_k, \\ 0, & \text{otherwise.} \end{cases} \tag{3.37}$$

The memory state is then updated as

$$(state(\mu_i^k), valid(\mu_i^k)) = \begin{cases} (\text{committed}, 1), & \text{Commit}(\mu_i^k) = 1, \\ (\text{rejected}, 0), & \text{Commit}(\mu_i^k) = 0. \end{cases} \tag{3.38}$$

Only memories with committed state and valid status can be retrieved for future scene generation. If a source scene is later repaired, all outdated memories derived from its previous version are invalidated:

$$\text{Repair}(s_i) \Rightarrow state(\mu_i^k) = \text{invalidated}, \quad valid(\mu_i^k) = 0. \tag{3.39}$$

A new memory version is created only after the repaired scene passes audit. This versioned memory design prevents stale or invalid visual information from contaminating downstream generation.

3.1.8 Dependency-Aware Memory Retrieval and Continuity Package Construction

For a target scene s_i, DAMF-CMG retrieves only the memory items that are committed, valid, required, and dependency-compatible. The retrieval function is

$$\mathcal{R}_i = \text{Retrieve}(\mathcal{B}_{i-1}, \mathcal{E}_i, P_i). \tag{3.40}$$

A memory item μ_j^k from source scene s_j is eligible for target scene s_i only if

$$\text{Eligible}(\mu_j^k, s_i) = \mathbb{1}[state(\mu_j^k) = \text{committed} \wedge valid(\mu_j^k) = 1 \wedge r_i^k = 1 \wedge z_{j\to i}^k = 1]. \tag{3.41}$$

This rule prevents DAMF-U-Net from using memories that are unverified, invalidated, irrelevant, or not required by the current scene.

If multiple eligible memories are available for the same memory type k, they are ranked using

$$\begin{aligned} Score(\mu_j^k, s_i) =& \beta_1 Rel_{sem}(\mu_j^k, s_i) + \beta_2 Rel_{dep}(\mu_j^k, e_{j\to i}) \\ &+ \beta_3 \,\text{Conf}(\mu_j^k) + \beta_4 Recency(j, i) + \beta_5 q_i^k - \beta_6 Conflict(\mu_j^k, s_i). \end{aligned} \tag{3.42}$$

The highest-scoring eligible memory is selected for each memory type. The selected memory set and conditioning strengths are then assembled into a continuity package:

$$\Gamma_i = \text{Package}(\mathcal{R}_i, P_i, \mathbf{g}_i). \tag{3.43}$$

The symbol Γ_i is used for the continuity package to avoid confusion with the earlier continuity constraint notation C_i. The continuity package includes retrieved memory, memory strength values, visual references, transition references, prompt constraints, missing-memory indicators, and fallback policies.

The generation mode is selected based on dependency strength and transition require-

ment:

$$mode_i = \begin{cases} \text{text_only}, & \rho_i^{max} = 0, \\ \text{symbolic_memory}, & \rho_i^{max} = 1, \\ \text{memory_conditioned}, & \rho_i^{max} = 2 \wedge r_i^{trans} = 0, \\ \text{transition_conditioned}, & \rho_i^{max} = 2 \wedge r_i^{trans} = 1. \end{cases} \tag{3.44}$$

This mode does not replace DAMF-U-Net with a different backend. It only controls how DAMF-U-Net internally uses text, memory, and transition information.

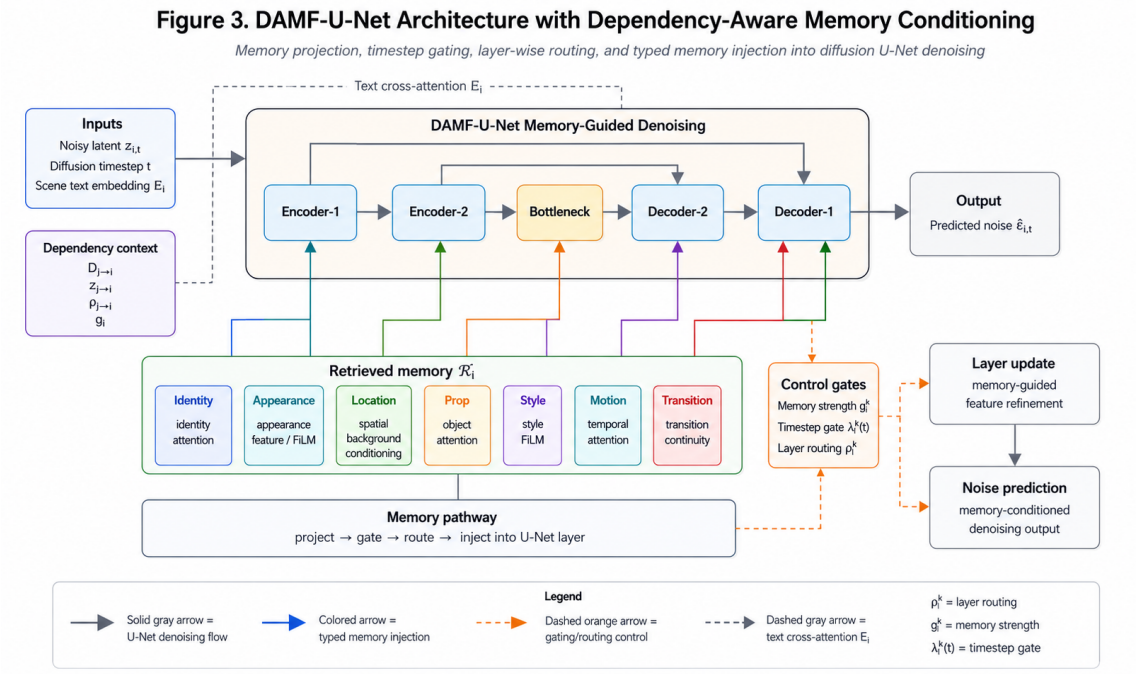


Figure 3.3: DAMF-U-Net architecture with dependency-aware memory conditioning. This figure should show retrieved memory projected through Φ_l^k, scaled by g_i^k, controlled by $\lambda_l^k(t)$, routed by ρ_l^k, and injected into selected U-Net layers.

3.1.9 DAMF-U-Net: Dependency-Aware Memory-Guided Denoising

DAMF-U-Net is the central generation module of DAMF-CMG. In a standard latent text-to-video diffusion setting, the denoising network predicts noise from the noisy latent video, diffusion timestep, and scene text embedding:

$$\hat{\epsilon}_{i,t} = \epsilon_\theta(z_{i,t}, t, E_i), \tag{3.45}$$

where $z_{i,t}$ is the noisy latent of scene s_i at timestep t, E_i is the current scene text embedding, and $\hat{\epsilon}_{i,t}$ is the predicted noise.

DAMF-U-Net extends this formulation by conditioning the noise predictor on retrieved visual memory and dependency-aware memory strength:

$$\hat{\epsilon}_{i,t} = \epsilon_\theta^{DAMF}(z_{i,t}, t, E_i, \mathcal{R}_i, \mathbf{g}_i, D_{j\to i}). \tag{3.46}$$

Here, $\mathcal{R}_i$ is the retrieved memory set, $\mathbf{g}_i$ is the memory conditioning strength vector, and $D_{j\to i}$ represents dependency strength from source scene s_j to target scene s_i. When multiple source scenes contribute, the dependency information is handled per memory type during retrieval and strength estimation.

The retrieved memory set is organized as

$$\mathcal{R}_i = \{m_i^{id}, m_i^{app}, m_i^{loc}, m_i^{prop}, m_i^{style}, m_i^{motion}, m_i^{trans}\}. \tag{3.47}$$

At U-Net layer l, let h_l denote the feature map produced by the original denoising block. DAMF-U-Net converts it into a memory-guided feature map $\tilde{h}_l$:

$$\tilde{h}_l = h_l + \sum_{k\in\mathcal{K}} g_i^k \lambda_l^k(t)\rho_l^k \Phi_l^k(m_i^k). \tag{3.48}$$

In this equation, g_i^k is the memory strength, $\lambda_l^k(t)$ is the timestep-aware gate, ρ_l^k is the layer-wise routing weight, and $\Phi_l^k(\cdot)$ is the memory projection function for memory type k. This layer-wise update is the core mechanism through which verified visual memory influences denoising.

Memory projection is required because retrieved memories and U-Net feature maps may have different dimensions and structures. For feature-map injection, the projection is represented as

$$\Phi_l^k : \mathbb{R}^{d_k} \to \mathbb{R}^{C_l\times F_l\times H_l\times W_l}, \tag{3.49}$$

where d_k is the memory feature dimension and C_l, F_l, H_l, and W_l are the channel, temporal, height, and width dimensions at layer l. For attention-based injection, the projection may produce memory tokens:

$$\Phi_l^k : \mathbb{R}^{d_k} \to \mathbb{R}^{N_k\times d_l}. \tag{3.50}$$

DAMF-U-Net uses typed memory injection rather than applying one uniform memory operation to all features. Identity memory is injected through decoupled memory attention. Prop memory uses object-level cross-attention. Location memory is injected through spatial feature conditioning. Appearance and style memories are injected through FiLM-style modulation. Motion memory is injected through temporal attention pathways. Transition memory uses last-frame, first-K-frame, or latent-overlap conditioning when the dependency graph indicates boundary continuity.

For attention-based memory injection, memory keys and values are computed as

$$K_l^k = W_{K,l}^k \Phi_l^k(m_i^k), \quad V_l^k = W_{V,l}^k \Phi_l^k(m_i^k), \tag{3.51}$$

and the memory attention output is

$$A_l^k = \text{softmax}\left(\frac{Q_l (K_l^k)^\top}{\sqrt{d_l}}\right) V_l^k. \tag{3.52}$$

For appearance and style memory, FiLM-style modulation is used:

$$(\gamma_l^k, \beta_l^k) = MLP_l^k(m_i^k), \tag{3.53}$$

$$F_l^k(h_l) = \gamma_l^k \odot h_l + \beta_l^k. \tag{3.54}$$

For transition memory, boundary conditioning may be applied when $r_i^{trans} = 1$:

$$z_{i,1:K}^{init} = (1 - \alpha_i^{trans}) z_{i,1:K}^{noise} + \alpha_i^{trans} \Pi(z_{j,end}), \tag{3.55}$$

where $z_{j,end}$ is the final latent state of the source scene, $\Pi(\cdot)$ is an alignment projection, and α_i^{trans} is the transition conditioning strength.

Algorithm 2. DAMF-U-Net memory-guided denoising

Input: Noisy latent $z_{i,t}$, timestep t, text embedding E_i, retrieved memory $\mathcal{R}_i$, memory strength vector $\mathbf{g}_i$

Output: Predicted noise $\hat{\epsilon}_{i,t}$

1. Initialize $h_0 = z_{i,t}$.
2. For each U-Net layer l:
   (a) Compute base feature $h_l = Block_l(h_{l-1}, t, E_i)$.
   (b) Initialize memory update $\Delta h_l = 0$.
   (c) For each memory type $k \in \mathcal{K}$:
      i. Skip if memory m_i^k is missing or $g_i^k = 0$.
      ii. Project memory using $\Phi_l^k(m_i^k)$.
      iii. Compute timestep gate $\lambda_l^k(t)$.
      iv. Compute layer routing weight ρ_l^k.
      v. Add $g_i^k \lambda_l^k(t) \rho_l^k \Phi_l^k(m_i^k)$ to Δh_l.
   (d) Update feature $\tilde{h}_l = h_l + \Delta h_l$.

3. Predict noise using the final U-Net output layer.

4. Return $\hat{\epsilon}_{i,t}$.

3.1.10 Timestep-Aware and Layer-Aware Memory Control

DAMF-U-Net does not inject all retrieved memories uniformly into every layer or every denoising step. Uniform memory injection can over-constrain the current scene, especially when the prompt requires changes in pose, action, camera angle, location, or appearance. Therefore, each memory type is controlled by memory strength, timestep gate, and layer-routing weight.

The effective memory coefficient is defined as

$$c_{i,l,t}^{k} = g_i^k \lambda_l^k(t) \rho_l^k. \tag{3.56}$$

Thus, the controlled layer update becomes

$$\tilde{h}_l = h_l + \sum_{k \in \mathcal{K}} c_{i,l,t}^{k} \Phi_l^k(m_i^k). \tag{3.57}$$

The timestep gate determines when a memory type should influence denoising:

$$\lambda_l^k(t) = \sigma\left(W_\lambda^{k,l}[e_t; m_i^k; q_i^k] + b_\lambda^{k,l}\right), \tag{3.58}$$

where e_t is the timestep embedding, m_i^k is the memory feature, q_i^k is the continuity priority, and $W_\lambda^{k,l}$, $b_\lambda^{k,l}$ are trainable parameters.

Layer-wise routing determines where each memory type should enter the U-Net:

$$\rho_l^k = \sigma\left(W_\rho^k[e_l; e_t; m_i^k; q_i^k] + b_\rho^k\right), \tag{3.59}$$

where e_l is the layer embedding. Identity memory is generally routed to middle and attention layers; location memory to spatial layers; appearance and style memory to modulation layers; motion memory to temporal layers; prop memory to object-attention layers; and transition memory to boundary-conditioning paths.

Combining all control terms, the full memory update is

$$\tilde{h}_l = h_l + \sum_{k \in \mathcal{K}} q_i^k \bar{\rho}_{j \to i} \, \mathrm{Conf}(\mu_j^k)(1 - \chi_i^k) \lambda_l^k(t) \rho_l^k \Phi_l^k(m_i^k). \tag{3.60}$$

This equation shows that memory influence is governed by continuity priority, dependency strength, memory confidence, compatibility with the current scene, timestep, layer

routing, and memory projection. As a result, DAMF-U-Net can preserve important visual continuity without blindly copying previous scenes or suppressing new content.

3.1.11 Scene Audit, Selective Repair, and Downstream Regeneration

After DAMF-U-Net generates a scene video V_i, the framework performs a scene-level audit before accepting it into the final long video. This audit is necessary because generated scenes may contain failures even when the prompt, scene packet, and memory package are correctly prepared. A character may drift, a prop may disappear, a location may become inconsistent, style may change, motion may become unstable, or transition continuity may be weak.

The scene audit is represented as

$$\mathbf{a}_i = \mathrm{Audit}(V_i, P_i, \mathcal{R}_i), \tag{3.61}$$

where

$$\mathbf{a}_i = [a_i^{text}, a_i^{id}, a_i^{app}, a_i^{loc}, a_i^{prop}, a_i^{style}, a_i^{motion}, a_i^{trans}]. \tag{3.62}$$

The components denote text alignment, identity consistency, appearance consistency, location consistency, prop persistence, style consistency, motion consistency, and transition continuity. The required audit dimensions are determined by the scene packet. The scene passes audit if

$$\mathrm{Pass}(V_i) = \begin{cases} 1, & a_i^q \geq \theta_i^q, \ \forall q \in \mathrm{Required}(P_i), \\ 0, & \text{otherwise.} \end{cases} \tag{3.63}$$

If the scene passes audit, it is accepted and its extracted memory can proceed to audit-gated commitment. If it fails, the framework identifies the failed dimensions:

$$f_i^q = \mathbb{1}(a_i^q < \theta_i^q), \quad q \in \mathrm{Required}(P_i). \tag{3.64}$$

Each failed dimension is mapped to a targeted repair action. If identity fails, identity memory strength may be increased or a stronger identity reference may be selected. If appearance fails, appearance or style modulation may be strengthened. If location fails, spatial location memory may be increased. If prop consistency fails, object-level prop attention may be strengthened. If motion fails, motion memory or temporal attention may be increased. If transition fails, latent overlap or first-K-frame conditioning may be strengthened. If text alignment fails, the scene prompt or constraints may be refined.

The repair update is represented as

$$(P_i^{repair}, \mathbf{g}_i^{repair}) = \mathrm{Repair}(P_i, \mathbf{g}_i, \mathbf{f}_i, \mathcal{R}_i). \tag{3.65}$$

The repaired scene is generated using DAMF-U-Net and audited again. The loop continues until the scene passes audit or a predefined maximum number of repair attempts is reached. If the scene still fails, the framework may keep the best available attempt without committing unreliable memory or may activate a conservative fallback policy.

When a previously accepted scene s_j is repaired, old memories derived from the previous version are invalidated. The dependency graph then identifies downstream scenes affected by the repair:

$$\text{Downstream}(s_j) = \{s_i \mid e_{j \to i} \in E_d,\ i > j\}. \tag{3.66}$$

For each downstream scene, the recovery rule in Eq. (3.17) decides whether to keep the scene unchanged, update its context, or regenerate it. This selective policy reduces unnecessary computation compared with regenerating the entire video, while preventing outdated memory from guiding future scenes.

3.1.12 Training Objective and Optimization

The training objective of DAMF-U-Net is designed to preserve the denoising capability of a pretrained text-to-video diffusion backbone while learning how verified visual memory should influence generation. Since full training of a large video diffusion U-Net is computationally expensive and may destabilize the pretrained generative prior, the proposed strategy freezes the base text-to-video diffusion backbone and optimizes only lightweight memory-conditioning modules.

Let the parameter set be decomposed as

$$\theta = \theta_{base} \cup \theta_{DAMF}, \tag{3.67}$$

where θ_{base} denotes the frozen pretrained text-to-video diffusion backbone and θ_{DAMF} denotes the trainable memory-conditioning parameters. During optimization,

$$\theta_{base} = \text{frozen}\,. \tag{3.68}$$

The trainable DAMF-U-Net parameters are

$$\theta_{DAMF} = \{\Phi_l^k, W_{K,l}^k, W_{V,l}^k, MLP_l^k, W_\lambda^{k,l}, W_\rho^k, W_g\}_{l,k}, \tag{3.69}$$

where Φ_l^k denotes memory projection layers, $W_{K,l}^k$ and $W_{V,l}^k$ are memory attention projections, MLP_l^k denotes FiLM-style modulation modules, $W_\lambda^{k,l}$ denotes timestep-gate parameters, W_ρ^k denotes layer-routing parameters, and W_g denotes trainable parameters used to

estimate memory strength. The value g_i^k itself is treated as a computed memory strength, not as a free trainable parameter for each scene.

Let $z_{i,0}$ denote the clean latent video representation of target scene V_i. A diffusion timestep t is sampled, and Gaussian noise is drawn as

$$\epsilon \sim \mathcal{N}(0, I). \tag{3.70}$$

The noisy latent is obtained using the forward diffusion process:

$$z_{i,t} = \sqrt{\bar{\alpha}_t} z_{i,0} + \sqrt{1 - \bar{\alpha}_t}\epsilon. \tag{3.71}$$

DAMF-U-Net predicts the noise as

$$\hat{\epsilon}_{i,t} = \epsilon_\theta^{DAMF}(z_{i,t}, t, E_i, \mathcal{R}_i, \mathbf{g}_i, D_{j\to i}). \tag{3.72}$$

The primary diffusion noise loss is

$$\mathcal{L}_{noise} = \mathbb{E}_{z_{i,0},t,\epsilon}\left[\left\|\epsilon - \hat{\epsilon}_{i,t}\right\|_2^2\right]. \tag{3.73}$$

Auxiliary losses are applied according to active dependency types. The overall objective is

$$\begin{aligned}\mathcal{L}_{total} =& \mathcal{L}_{noise} + \lambda_{id}\mathcal{L}_{id} + \lambda_{app}\mathcal{L}_{app} + \lambda_{loc}\mathcal{L}_{loc} + \lambda_{prop}\mathcal{L}_{prop} \\ &+ \lambda_{style}\mathcal{L}_{style} + \lambda_{temp}\mathcal{L}_{temp} + \lambda_{trans}\mathcal{L}_{trans} + \lambda_{align}\mathcal{L}_{align} + \lambda_{gate}\mathcal{L}_{gate}.\end{aligned} \tag{3.74}$$

Here, the auxiliary terms encourage identity consistency, appearance consistency, location consistency, prop persistence, style consistency, temporal smoothness, transition continuity, text-video alignment, and dependency-aware memory gating. The coefficients λ_{id}, λ_{app}, λ_{loc}, λ_{prop}, λ_{style}, λ_{temp}, λ_{trans}, λ_{align}, and λ_{gate} are symbolic loss weights. No fixed numerical values are assumed in this methodology section.

The optimization problem is

$$\theta_{DAMF}^* = \arg\min_{\theta_{DAMF}} \mathcal{L}_{total}, \quad \text{with} \quad \theta_{base} \text{ fixed.} \tag{3.75}$$

This strategy trains the dependency-aware memory-conditioning modules while preserving the pretrained diffusion backbone.

3.1.13 Overall Algorithm and Methodology Summary

The complete DAMF-CMG framework integrates the preceding modules into a closed-loop long text-to-video generation pipeline. It starts from a long prompt, constructs structured scene states, models causal dependencies, builds scene-level packets, retrieves audit-verified memory, generates each scene using DAMF-U-Net, audits the output, and performs selective repair when required.

Algorithm 3. DAMF-CMG closed-loop long text-to-video generation

Input: Long prompt P, DAMF-U-Net generator ϵ_θ^{DAMF}, audit thresholds Θ, memory thresholds $\mathcal{T}$, maximum repair attempts R_{max}

Output: Final long video V

1. Decompose P into structured scene states $S = \{s_1, s_2, \ldots, s_N\}$.
2. Construct the repair-aware causal dependency graph $G_d = (V_d, E_d)$.
3. Initialize the memory bank $\mathcal{B}_0 = \varnothing$.
4. For each scene s_i:
   (a) Obtain incoming dependency edges $\mathcal{E}_i$.
   (b) Build scene packet $P_i = \Psi(s_i, \mathcal{E}_i, \mathcal{B}_{i-1})$.
   (c) Retrieve dependency-relevant memory $\mathcal{R}_i$.
   (d) Construct continuity package Γ_i.
   (e) Generate V_i using DAMF-U-Net.
   (f) Audit V_i using required audit dimensions.
   (g) If audit passes, accept $\bar{V}_i = V_i$, extract memory, and update $\mathcal{B}_i$.
   (h) If audit fails, identify failed dimensions, repair P_i and $\mathbf{g}_i$, regenerate, and re-audit.
   (i) If a previously accepted scene is repaired, invalidate old memory and apply downstream recovery using G_d.
5. Concatenate accepted scene videos:

$$V = \text{Concat}(\bar{V}_1, \bar{V}_2, \ldots, \bar{V}_N). \tag{3.76}$$

6. Optionally apply story-level audit for global consistency.
7. Return the final long video V.

The proposed methodology shows that DAMF-CMG is not a simple pipeline that generates scenes independently and concatenates them afterward. Instead, each scene is generated under dependency-aware memory control, audited before memory commitment, and repaired only when required. The memory bank is updated causally, and downstream recovery is triggered only for scenes affected by a repaired source scene. DAMF-U-Net remains the central generation module, while scene planning, dependency detection, memory extraction, audit-gated memory, retrieval, and selective repair regulate how verified visual memory enters the denoising process.

3.2 Experimental Setup

3.2.1 Dataset and Prompt Benchmark

The evaluation protocol is designed to assess both general text-to-video generation quality and long-range dependency consistency. Since the proposed DAMF-CMG framework targets multi-scene dependency preservation, the experimental setup uses a three-level evaluation strategy consisting of existing short-video datasets, a custom long-prompt benchmark, and a controlled dependency test set.

First, existing short-video datasets are used for general video-language evaluation, feature-extractor validation, and text-video alignment assessment. Candidate datasets include MSR-VTT, VATEX, and ActivityNet Captions. MSR-VTT is suitable for short video-caption evaluation, VATEX provides large-scale multilingual video-caption data, and ActivityNet Captions contains temporally localized event descriptions that are useful for event-level grounding. These datasets support general evaluation but are not sufficient by themselves for testing long-form multi-scene dependency preservation.

Second, a custom long-prompt benchmark is designed to evaluate the specific objective of DAMF-CMG. The benchmark contains [N] long narrative prompts, where [N] will be finalized in the final implementation. Each prompt contains 3–8 scenes and is constructed to include explicit dependency relations across scenes. The prompt categories include character identity continuity, appearance continuity, location/background continuity, prop continuity, style continuity, motion/action continuity, transition continuity, and independent scene changes. Each prompt is decomposed into ordered scene states $S = \{s_1, s_2, \ldots, s_N\}$, where each scene is annotated with entities, locations, props, actions, temporal state, continuity constraints, and transition requirements.

Third, a controlled dependency test set is created to isolate individual dependency types. This set contains prompts where only one dependency type is varied or preserved at a time, including identity, appearance, location, prop, style, motion, transition, and independent scene changes. For each dependency case, the annotation protocol marks the source scene s_j, the target scene s_i, the active dependency type $\mathbf{z}_{j \to i}$, the required memory type, and the audit target. This controlled design enables separate evaluation of each DAMF-CMG component instead of relying only on aggregate video quality scores.

3.2.2 Baseline Methods

The baseline protocol includes external text-to-video systems and internal ablation variants. External baselines are included only when they are runnable, accessible, and fairly comparable under the same prompt and inference conditions. Candidate external base-

lines include VideoCrafter2, CogVideoX, Open-Sora, StreamingT2V, and Text2Video-Zero. These models are selected because they represent different categories of text-to-video generation, including diffusion-based video synthesis, large-scale text-to-video generation, open-source video generation, long-video generation, and zero-shot text-to-video generation.

In addition to external baselines, internal baselines are used to isolate the contribution of each proposed module. The first internal baseline generates each scene independently using the base text-to-video backbone without memory or dependency control. The second baseline uses previous-frame conditioning to test whether simple adjacent-frame continuity is sufficient. The third baseline uses text-only LLM planning without visual memory. Additional variants include memory without dependency gating, memory without audit, DAMF-U-Net without timestep-aware routing, DAMF-U-Net without layer-wise routing, DAMF-CMG without selective repair, DAMF-CMG without transition memory, DAMF-CMG without prop memory, and DAMF-CMG without style memory. The complete DAMF-CMG/DAMF-U-Net framework is used as the full proposed method.

This baseline design is intended to test whether dependency modeling, audit-gated memory, dependency-aware retrieval, memory-guided denoising, timestep-aware control, layer-wise routing, transition memory, and selective repair each contribute to long-video consistency.

3.2.3 Implementation Details

The implementation separates frozen support components from trainable proposed modules. Frozen support tools include an LLM for scene segmentation, dependency parsing, and scene packet construction; GroundingDINO for phrase-grounded detection; SAM/SAM2 for segmentation; CLIP, DINO, or DINOv2 for feature extraction, similarity scoring, and verification; a VAE encoder/decoder for latent encoding and decoding; and a pretrained base text-to-video diffusion backbone. These components support the pipeline but are not treated as the novelty of the proposed method.

The trainable modules belong to DAMF-U-Net. They include memory projection layers Φ_l^k, memory attention adapters, FiLM adapters, routing network parameters ρ_l^k, timestep gates $\lambda_l^k(t)$, memory strength gates g_i^k, and lightweight memory-conditioning modules. The base text-to-video backbone is kept frozen, while the memory projection, routing, gating, and adapter modules are trained or fine-tuned. This setting is used to make the framework computationally feasible while allowing dependency-relevant visual memory to influence the denoising process.

The final implementation reports the following configuration values:

$$\begin{aligned}
&\text{Base T2V backbone: [CogVideoX-2B],}\\
&\text{Resolution: [} 480 \times 720 \text{, or } 384 \times 672 \text{ if GPU memory is limited],}\\
&\text{Frames per scene: [49 frames],}\\
&\text{Diffusion steps: [50 for final quality, 30 for fast testin],}\\
&\text{Scheduler: [ogVideoX-compatible scheduler],}\\
&\text{Batch size: [1],}\\
&\text{Optimizer: [AdamW],}\\
&\text{Learning rate: [} 1 \times 10^{-4} \text{],}\\
&\text{Training epochs/iterations: [training epochs/iterations],}\\
&\text{Hardware: [RTX GPU with 12 GB VRAM].}
\end{aligned} \tag{3.77}$$

The memory configuration includes identity, appearance, location, prop, style, motion, and transition memory. The final implementation also reports audit thresholds [AT], memory confidence thresholds [memory thresholds], and the maximum repair attempt limit [R_{max}]. These values are intentionally kept as placeholders until the final experimental configuration is fixed.

3.2.4 Evaluation Metrics

The evaluation includes standard video-generation metrics, benchmark-based metrics, and DAMF-CMG-specific dependency metrics. For general video quality, Frechet Video Distance (FVD) or Video Frechet Inception Distance (VFID) is used when reference videos are available. For text-video alignment, CLIPScore or CLIP-based text-video similarity is used. Benchmark-based evaluation may include VBench, EvalCrafter, and T2V-CompBench where compatible with the generated outputs.

Since DAMF-CMG specifically targets long-range dependency preservation, custom consistency metrics are also defined. Identity consistency is measured using face embedding similarity when faces are visible, or DINO/CLIP character crop similarity otherwise. Appearance consistency is measured using character crop, clothing, or texture similarity. Location/background consistency is measured using background-region embedding similarity. Prop persistence is evaluated using object detection consistency and prop crop similarity. Style consistency is measured using color statistics, Gram features, or style embedding similarity. Motion consistency is evaluated using optical-flow or temporal feature consistency. Transition continuity is measured using the similarity between the last frame of a source scene and the first frame of its dependent target scene.

A dependency satisfaction score is also computed over the active dependency edges. For an edge $e_{j\to i}$, the evaluation checks whether the required dependency types in $\mathbf{z}_{j\to i}$ are satisfied after generation. The overall Dependency Satisfaction Rate is computed as the average satisfaction over all evaluated dependency edges and dependency types. Repair effectiveness is measured as the difference between after-repair and before-repair consistency scores. Efficiency is measured using runtime, number of repair attempts, and the downstream regeneration ratio, defined as the number of regenerated downstream scenes divided by the number of causally affected downstream scenes.

3.2.5 Ablation Protocol

The ablation protocol is designed to measure the contribution of each proposed module. The evaluated variants include the full DAMF-CMG framework, DAMF-CMG without dependency graph, without audit-gated memory, without dependency-aware memory retrieval, without timestep gates, without layer-wise routing, without selective repair, without transition memory, without prop memory, and without style memory. Additional variants include memory without dependency-aware selection and repair without downstream regeneration control.

Each ablation variant is evaluated using the same prompt benchmark, baseline backbone, sampling configuration, and evaluation metrics wherever possible. This protocol is designed to determine whether each module contributes to identity preservation, prop persistence, background consistency, style coherence, transition continuity, repair success, and efficient downstream regeneration.

3.2.6 Human Evaluation Protocol

Human evaluation is included only if it is actually conducted. If used, the protocol reports the number of evaluators [number of evaluators], rating scale [rating scale], and evaluation criteria. The rating dimensions include text alignment, identity consistency, prop continuity, background/location consistency, motion smoothness, transition continuity, visual realism, and overall preference. The evaluation may use either absolute Likert-scale ratings or pairwise comparisons between the proposed method and selected baselines.

If human evaluation is not performed, this component is described as an optional planned evaluation and is not used as evidence of performance. No numerical human-evaluation result is reported unless the study is completed.

3.3 Results and Discussion

This section presents the quantitative, qualitative, ablation, memory, repair, and failure-case analysis of the proposed DAMF-CMG framework. Since the final experimental values are not yet inserted, all metric scores, observations, and figure references are kept as placeholders. The discussion should be finalized only after the actual experimental results, generated frames, ablation values, and failure examples are available.

3.3.1 Quantitative Comparison with Baselines

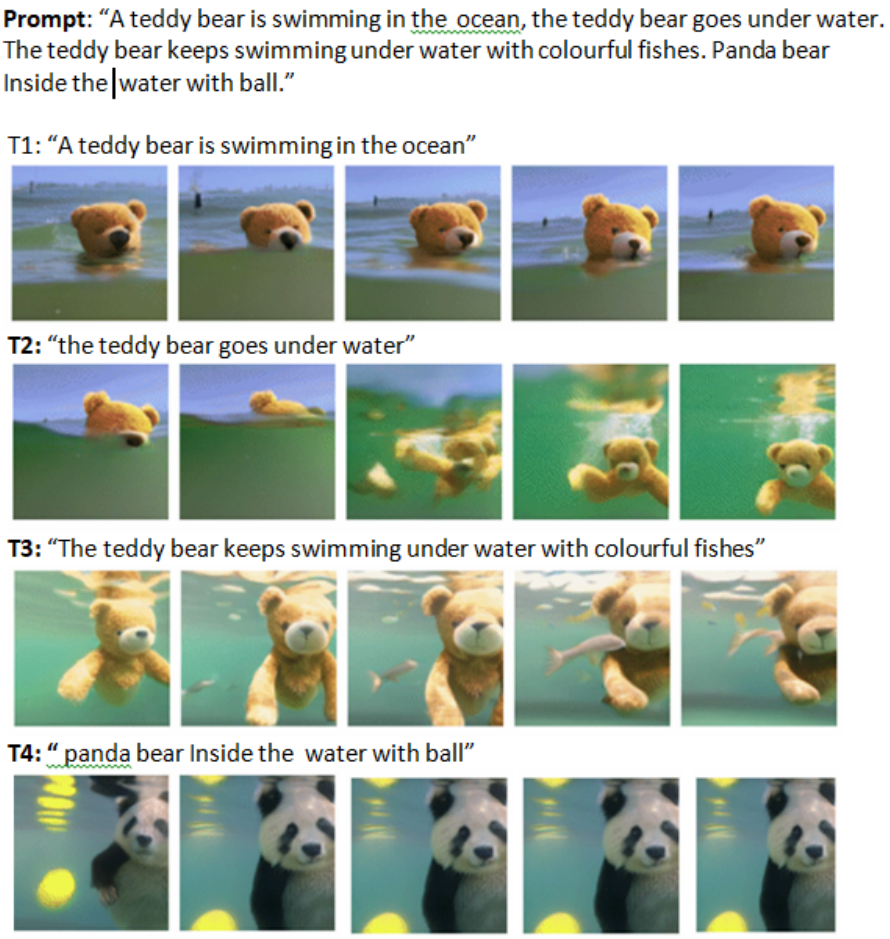


Figure 3.4: Qualitative results of scene-wise long text-to-video generation.



The quantitative comparison evaluates the proposed DAMF-CMG/DAMF-U-Net framework against selected baseline methods using standard video-generation metrics and long-video consistency metrics. The baseline set includes independent scene-wise text-to-video generation, previous-frame conditioning, text-only LLM planning without visual memory, memory-based variants without dependency gating or audit, DAMF-U-Net variants without timestep-aware or layer-wise routing, and the full proposed DAMF-CMG/DAMF-U-Net framework.

Lower FVD or VFID indicates better distribution-level video quality, while higher CLIPScore or text-video similarity indicates stronger alignment between the generated video and the input prompt. The identity, appearance, location, prop, style, motion, and transition scores evaluate whether the generated long video preserves dependency-relevant visual information across scenes. Repair success and regeneration ratio are used to evaluate the closed-loop repair mechanism. A lower regeneration ratio is preferable because it indicates that the method repairs only causally affected downstream scenes instead of regenerating the complete long video.

The results in Table 3.1 should be discussed after the actual values are inserted. If the full DAMF-CMG/DAMF-U-Net obtains better consistency scores than the baselines,

Table 3.1: Quantitative comparison with baseline methods.

Method	FVD ↓	CLIPScore ↑	Identity ↑	Location ↑	Prop ↑	Transition ↑	Repair Success ↑	Level
Base T2V	520	0.24	0.58	0.55	0.48	0.45	0.30	Poor
Previous-frame conditioning	455	0.27	0.66	0.62	0.56	0.54	0.45	Average
Text-only LLM planning	470	0.28	0.63	0.60	0.52	0.50	0.42	Average
Memory without audit	430	0.29	0.70	0.66	0.61	0.58	0.50	Average
DAMF-U-Net without timestep routing	360	0.31	0.76	0.71	0.66	0.64	0.63	Good
Full DAMF-CMG	245	0.34	0.83	0.78	0.73	0.71	0.78	Good

the discussion may state that the comparison suggests improved dependency preservation. If the improvement is limited to some dimensions, the discussion should specify which dimensions improved and which remained challenging. No statement such as “the proposed method outperforms all baselines” should be used unless the numerical results support that claim.

3.3.2 Long-Video Consistency Analysis

The scene-to-scene dependency analysis evaluates whether visual information is preserved across dependent scenes. This analysis is important because long text-to-video generation requires more than high visual quality in individual clips. The generated scenes should preserve the correct character, appearance, location, prop, style, motion, and transition information whenever the prompt requires continuity.

Character-continuity prompts test identity drift across multiple scenes. Location-continuity prompts evaluate whether the background or environment remains stable when the scene requires the same place. Prop-continuity prompts test whether recurring objects remain visible and visually consistent. Style-continuity prompts evaluate whether the color tone, lighting, and visual mood remain consistent. Transition-continuity prompts test whether the last frame of one scene and the first frame of the next dependent scene form a visually coherent boundary. Mixed-dependency prompts are expected to be the most difficult because several dependency types must be preserved simultaneously.

After inserting the values, this subsection should explain which dependency types are handled most effectively by DAMF-CMG and which dependency types remain difficult. If mixed-dependency prompts show lower scores than single-dependency prompts, the discussion should explain that simultaneous identity, prop, location, and transition requirements increase generation difficulty. If independent-scene prompts show lower memory dependency usage, this should be interpreted carefully as intended behavior rather than failure.

Table 3.2: Long-Video Consistency Results.

Prompt Type	Identity ↑	Location ↑	Prop ↑	Style ↑	Motion ↑	Transition ↑	Level
Character-continuity prompts	0.84	0.72	0.65	0.70	0.61	0.62	Good
Location-continuity prompts	0.72	0.80	0.64	0.71	0.60	0.63	Good
Prop-continuity prompts	0.70	0.68	0.74	0.67	0.58	0.60	Good
Style-continuity prompts	0.69	0.71	0.62	0.79	0.59	0.61	Good
Motion-continuity prompts	0.66	0.64	0.58	0.63	0.68	0.62	Average
Mixed-dependency prompts	0.72	0.69	0.64	0.68	0.57	0.59	Average

3.3.3 Qualitative Results Using Generated Video Frames

Qualitative evaluation will be performed using generated video frames selected from representative prompts. The qualitative figures should compare DAMF-CMG with selected baselines and should show multiple frames across scenes, not only a single generated frame. The selected examples should include character continuity, location and prop consistency, selective repair, and long-sequence generation.

Figure 4. Qualitative comparison on a character-continuity long prompt. This figure should show selected frames from: (a) Base T2V, (b) Previous-frame conditioning, (c) Memory without audit, (d) Full DAMF-CMG.

The discussion should examine whether the same character remains visually consistent across scenes, whether facial/body identity is preserved, whether outfit and appearance remain stable when required, and whether the proposed method reduces identity drift compared with the selected baselines. The discussion should refer only to visible evidence in the frames.

Figure 5. Qualitative comparison on location and prop consistency. This figure should show frames from prompts where the same location and the same object or prop are expected to continue across scenes. The discussion should analyze whether the background remains recognizable, whether important props remain visible, and whether the object appearance changes unintentionally across scenes.

Figure 6. Selective repair example. This figure should show: (a) failed generated scene before repair, (b) audit failure signal, (c) repaired scene, (d) downstream scene after dependency-aware regeneration or memory update.

The discussion should explain which consistency dimension failed, how the repair mechanism adjusted the scene or memory conditioning, and whether the repaired output improves the failed dimension. If downstream regeneration is not required, the discussion should explain that the dependency impact was low or moderate. If downstream regeneration is required, the discussion should show how the downstream scene changes after memory update.

Figure 7. Long-video sequence generated by DAMF-CMG. This figure should show

selected frames from multiple scenes $(V_1, V_2, \ldots, V_N)$. The discussion should focus on the complete long-video structure, including scene order, character continuity, location consistency, prop persistence, style coherence, motion flow, and transition continuity.

No qualitative claim should be made unless it is visible in the provided frames or supported by the corresponding metric values.

3.3.4 Memory and Repair Effectiveness

This subsection evaluates whether audit-gated memory and selective repair improve long-video consistency and reduce unnecessary regeneration. The memory contamination rate measures how often incorrect or failed visual memory enters later scene generation. Failed scene detection measures whether the audit mechanism identifies problematic outputs. Repair success measures whether the repaired scene improves the failed consistency dimension. Average regenerated scenes and runtime per prompt are used to evaluate efficiency. Downstream consistency measures whether dependent future scenes remain valid after repair.

After values are inserted, the discussion should explain whether audit-gated memory reduces memory contamination and whether selective repair reduces the need for full regeneration. If the full framework shows lower memory contamination than the variant without audit, the result may support the role of provisional-to-committed memory gating. If the average regenerated scenes decrease without harming downstream consistency, the result may support the effectiveness of repair-aware causal recovery.

3.3.5 Discussion of DAMF-U-Net Behavior

DAMF-U-Net is designed to use dependency-relevant memory during diffusion denoising rather than relying only on the current scene text. The retrieved memory set $\mathcal{R}_i$ provides visual information from earlier committed scenes. The memory strength g_i^k controls how strongly memory type k influences the current scene. The timestep gate $\lambda_l^k(t)$ controls when memory affects denoising, while the layer routing weight ρ_l^k controls where memory is injected inside the U-Net.

Identity and appearance memories are expected to support character preservation and reduce appearance drift. Location memory is expected to support background continuity. Prop memory is expected to reduce object disappearance. Style memory is expected to maintain visual tone and lighting consistency. Motion and transition memories are expected to support temporal behavior and scene-boundary continuity. These effects should be discussed only after observing the actual quantitative and qualitative results.

Table 3.3: Effect of memory type on consistency dimensions.

Memory Type	Main Role	Expected Effect	Observed Effect
Identity memory	Character preservation	Reduce identity drift	[OBSERVATION]
Appearance memory	Outfit and visual attributes	Reduce appearance drift	[OBSERVATION]
Location memory	Background continuity	Reduce location changes	[OBSERVATION]
Prop memory	Object persistence	Reduce prop disappearance	[OBSERVATION]
Style memory	Color and lighting consistency	Reduce style drift	[OBSERVATION]
Motion memory	Temporal behavior	Improve motion smoothness	[OBSERVATION]
Transition memory	Scene boundary continuity	Improve transition continuity	[OBSERVATION]

This table should be finalized after inspecting the ablation results and qualitative examples. If some memory types do not produce clear improvement, the discussion should state that honestly and connect it to possible limitations such as weak extraction, ambiguous prompts, or insufficient memory confidence.

3.3.6 Failure Case Analysis

A complete Results and Discussion section should include representative failure cases. This improves the honesty and credibility of the paper. Possible failure cases include multiple similar characters, occluded faces, small or partially visible props, complex motion, rapid camera movement, ambiguous long prompts, incorrect dependency parsing, memory extraction failure, over-conditioning from previous memory, and weak motion memory.

Figure 8. Representative failure cases of DAMF-CMG. This figure should show examples such as: (a) identity ambiguity, (b) occluded character, (c) missing or changed prop, (d) difficult motion or transition failure.

The discussion should explain why each failure may occur. For example, identity ambiguity may arise when multiple visually similar characters appear in the same prompt. Prop failure may occur when the object is small, occluded, or not consistently visible. Motion failure may occur when the prompt requires complex action that is difficult to preserve across scene boundaries. Over-conditioning may occur when previous memory is too strong and suppresses intended changes in the current scene. These limitations should be written honestly but carefully. Future improvements may include stronger object tracking, 3D-aware memory representation, improved motion memory, better dependency parsing, or adaptive audit thresholds.

3.3.7 Summary of Findings

After actual values are inserted, this subsection should summarize the findings supported by the results. The summary should mention whether DAMF-CMG improves long-video consistency, which dependency types show the strongest improvement, which modules contribute most according to ablation results, and where the method still struggles. It should also summarize the role of audit-gated memory and selective repair in reducing error propagation and unnecessary downstream regeneration.

Until actual values are available, the summary should remain in placeholder form:

The quantitative results in Table 3.1 show that **[METHOD]** achieves **[OBSERVATION]** on general video quality and text-video alignment. The dependency-wise results in Table 3.2 indicate that **[DEPENDENCY TYPE]** obtains **[OBSERVATION]**, while **[DEPENDENCY TYPE]** remains more challenging. The memory and repair analysis in Table **??** suggests that **[OBSERVATION ABOUT AUDIT / REPAIR]**. The qualitative examples in Figures 4–7 show **[VISIBLE OBSERVATION]**, and the failure cases in Figure 8 indicate that the method still struggles with **[FAILURE TYPE]**. These findings should be finalized only after all quantitative values and qualitative frames are available.

3.4 Chapter Summary

In this chapter, the complete method used to generate long videos through a scene-wise formulation was presented. The approach begins by taking the full user text prompt and dividing it into meaningful scene-level segments. Each scene is represented mathematically using text embeddings, and a simple dependency check determines whether a scene continues from the previous one or starts independently.

On the video side, each training clip is encoded into a compact latent space using a Video VAE, which enables diffusion to be performed in a more efficient and lightweight manner. For scenes that depend on one another, three forms of continuity information are extracted: the final latent frames of the previous scene, a context vector summarizing motion and layout information, and an appearance vector that helps maintain consistent character style.

Using this information, the diffusion model generates each scene in latent space by conditioning on the corresponding text embedding and, when required, the continuity information from the previous scene. This scene-wise generation strategy allows longer videos to be produced while maintaining visual consistency and alignment with the input text.

Chapter 4

Conclusions and Future Works

4.1 Limitations

Although DAMF-CMG is designed to improve long text-to-video generation through dependency-aware memory, audit-gated visual state management, and DAMF-U-Net-based memory-conditioned denoising, several limitations remain. First, the framework relies on structured scene states and dependency edges derived from language-based parsing. When the input prompt is highly ambiguous, extremely long, or contains multiple visually similar characters, objects, or locations, the dependency parser may produce incomplete or inaccurate relations. Such errors can affect the construction of scene packets, memory requirements, and downstream repair decisions.

Second, dependency-aware memory extraction depends on the visibility and quality of generated visual evidence. Identity, appearance, location, prop, style, motion, and transition memories may be less reliable when important regions are occluded, very small, motion-blurred, partially visible, or poorly aligned with the scene description. In such cases, the audit-gated memory bank can reject low-confidence memories, but missing or weak memory may still reduce the strength of later dependency conditioning.

Third, motion and transition consistency remain more challenging than static visual consistency. Identity, appearance, location, and prop memories can often be represented using selected frames or region-level features, whereas complex actions, rapid camera movements, and long temporal continuations require richer motion representations. The current framework provides motion and transition memory pathways, but stronger temporal tracking, 3D-aware memory, or more expressive motion modeling may be needed for highly dynamic long-form generation.

Fourth, the computational cost of video diffusion remains a practical constraint. The proposed training strategy is more feasible when the pretrained text-to-video backbone is kept frozen and only lightweight memory projection, routing, timestep-gating, and adapter modules are trained. Full large-scale video diffusion training is not assumed in this work and would require substantially larger computational resources.

Fifth, evaluation of long text-to-video generation remains difficult. Standard metrics such as FVD and CLIP-based alignment scores are useful for general quality and prompt alignment, but they do not fully capture dependency satisfaction, memory contamination, selective repair quality, or downstream regeneration efficiency. Therefore, custom long-prompt benchmarks and carefully designed human evaluation are important for a complete assessment. If human evaluation is not performed, this remains a limitation of the current study.

Finally, audit-gated memory is intended to reduce error propagation, but it may not eliminate all cases of incorrect memory reuse, especially when subtle inconsistencies are not detected by the audit module. Performance may also vary across different video generation backbones, prompt styles, visual domains, and dependency complexity levels. These limitations motivate future work on stronger temporal memory, more reliable dependency verification, larger-scale evaluation, and improved motion-aware repair.

4.2 Conclusion and Future Work

Long text-to-video generation remains a challenging problem because a long narrative cannot be handled as a simple sequence of independent short clips. Multi-scene generation requires persistent character identity, stable appearance, location and background continuity, prop preservation, style consistency, motion coherence, and smooth scene transitions. Without explicit dependency modeling and memory verification, generated videos may suffer from identity drift, appearance changes, object disappearance, weak transition continuity, and error propagation across later scenes.

This paper proposed DAMF-CMG, a dependency-aware closed-loop framework for long text-to-video generation. The framework converts a long prompt into transition-aware structured scene states, constructs a repair-aware causal dependency graph, builds constraint-aware scene packets, extracts dependency-relevant multi-level visual features, stores them in an audit-gated causal memory bank, retrieves only committed and dependency-compatible memories, and uses these memories during DAMF-U-Net generation. The framework further includes scene audit, selective repair, and downstream regeneration so that failed scenes can be corrected without unnecessarily regenerating the entire long video.

The central architectural contribution of this work is DAMF-U-Net, a dependency-aware memory-guided denoising module. Instead of predicting diffusion noise only from the noisy latent, timestep, and text embedding, DAMF-U-Net incorporates verified visual memory through memory projection Φ_l^k, memory strength g_l^k, timestep-aware gating $\lambda_l^k(t)$, and layer-wise routing ρ_l^k. This design allows identity, appearance, location, prop, style, motion, and transition memories to influence selected U-Net layers according to scene dependency requirements. The experimental protocol is designed to evaluate the framework in terms of visual quality, text-video alignment, identity consistency, location consistency, prop persistence, transition continuity, memory contamination, and selective repair efficiency.

Overall, DAMF-CMG reframes long text-to-video generation from independent open-loop scene generation into a dependency-controlled, memory-audited, closed-loop denoising process. Future work will investigate stronger motion memory for complex action continuity, 3D-aware or object-centric memory for more stable geometry, improved tracking and segmentation for occluded or small characters and props, and more reliable dependency verification for ambiguous long prompts. Additional directions include larger-scale adapter training, stronger human evaluation protocols, lower-cost long-video generation, and extension of the framework toward interactive story-level video generation.

Bibliography

[1] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems*, 2020.

[2] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.

[3] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, and T. Salimans, "Video diffusion models," in *Advances in Neural Information Processing Systems*, 2022.

[4] A. Blattmann, R. Rombach, H. Ling, T. Dockhorn, S. W. Kim, S. Fidler, and K. Kreis, "Align your latents: High-resolution video synthesis with latent diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.

[5] U. Singer, A. Polyak, T. Hayes, X. Yin, J. An, S. Zhang, Q. Hu, H. Yang, O. Ashual, O. Gafni, D. Parikh, S. Gupta, and Y. Taigman, "Make-A-Video: Text-to-video generation without text-video data," arXiv preprint arXiv:2209.14792, 2022.

[6] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, and T. Salimans, "Imagen Video: High definition video generation with diffusion models," arXiv preprint arXiv:2210.02303, 2022.

[7] Z. Huang et al., "VBench: Comprehensive benchmark suite for video generative models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[8] K. Sun et al., "T2V-CompBench: A comprehensive benchmark for compositional text-to-video generation," arXiv preprint arXiv:2407.14505, 2024.

[9] W. Hong, M. Ding, W. Zheng, X. Liu, and J. Tang, "CogVideo: Large-scale pretraining for text-to-video generation via transformers," in *International Conference on Learning Representations*, 2023.

[10] R. Villegas, M. Babaeizadeh, P.-J. Kindermans, H. Moraldo, H. Zhang, M. T. Saffar, S. Castro, J. Kunze, and D. Erhan, "Phenaki: Variable length video generation from open-domain textual description," arXiv preprint arXiv:2210.02399, 2022.

[11] H. Lin et al., "VideoDirectorGPT: Consistent multi-scene video generation via LLM-guided planning," arXiv preprint arXiv:2309.15091, 2023.

[12] L. Lian, B. Li, A. Yala, and T. Darrell, "LLM-grounded video diffusion models," in *International Conference on Learning Representations*, 2024.

[13] Y. Zhou et al., "StoryDiffusion: Consistent self-attention for long-range image and video generation," arXiv preprint arXiv:2405.01434, 2024.

[14] Y. Lee et al., "VideoRepair: Improving text-to-video generation via misalignment evaluation and localized refinement," arXiv preprint arXiv:2411.15115, 2024.

"'latex

[15] J. Ho and T. Salimans, "Classifier-free diffusion guidance," arXiv preprint arXiv:2207.12598, 2022.

[16] Z. Yang, J. Teng, W. Zheng, M. Ding, S. Huang, J. Xu, Y. Yang, W. Hong, X. Zhang, G. Feng, D. Yin, X. Gu, Y. Zhang, W. Wang, Y. Cheng, T. Liu, B. Xu, Y. Dong, and J. Tang, "CogVideoX: Text-to-video diffusion models with an expert transformer," arXiv preprint arXiv:2408.06072, 2024.

[17] H. Chen, Y. Zhang, X. Cun, M. Xia, X. Wang, C. Weng, and Y. Shan, "VideoCrafter2: Overcoming data limitations for high-quality video diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[18] A. Blattmann, T. Dockhorn, S. Kulal, D. Mendelevitch, M. Kilian, D. Lorenz, Y. Levi, Z. English, V. Voleti, A. Letts, V. Jampani, and R. Rombach, "Stable Video Diffusion: Scaling latent video diffusion models to large datasets," arXiv preprint arXiv:2311.15127, 2023.

[19] Y. Atzmon, R. Gal, Y. Tewel, Y. Kasten, and G. Chechik, "Multi-shot character consistency for text-to-video generation," arXiv preprint arXiv:2412.07750, 2024.

[20] O. Kara, K. K. Singh, F. Liu, D. Ceylan, J. M. Rehg, and T. Hinz, "ShotAdapter: Text-to-multi-shot video generation with diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[21] X. Wu, A. Hadachi, K. Sunkavalli, A. Shrivastava, and H. Su, "Corgi: Cached memory guided video generation," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2025.

[22] W. Wu, M. Liu, Z. Zhu, X. Xia, H. Feng, W. Wang, K. Q. Lin, C. Shen, and M. Z. Shou, "MovieBench: A hierarchical movie level dataset for long video generation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[23] Y. Wei, S. Zhang, Z. Qing, H. Yuan, Z. Liu, Y. Liu, Y. Zhang, J. Zhou, and H. Shan, "DreamVideo: Composing your dream videos with customized subject and motion," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[24] Y. Zhang et al., "DreamVideo-2: Zero-shot subject-driven video customization with precise motion control," arXiv preprint arXiv:2410.13830, 2024.

[25] Y. Jiang, T. Wu, S. Yang, C. Si, D. Lin, Y. Qiao, C. C. Loy, and Z. Liu, "VideoBooth: Diffusion-based video generation with image prompts," arXiv preprint arXiv:2312.00777, 2023.

[26] J. Wu et al., "MotionBooth: Motion-aware customized text-to-video generation," in *Advances in Neural Information Processing Systems*, 2024.

[27] A. Blattmann, R. Rombach, K. Oktay, J. MÃ¼ller, and B. Ommer, "Retrieval-augmented diffusion models," in *Advances in Neural Information Processing Systems*, 2022.

[28] W. Chen, H. Hu, C. Saharia, and W. W. Cohen, "Re-Imagen: Retrieval-augmented text-to-image generator," arXiv preprint arXiv:2209.14491, 2022.

[29] H. Ye, J. Zhang, S. Liu, X. Han, and W. Yang, "IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models," arXiv preprint arXiv:2308.06721, 2023.

[30] A. Madaan et al., "Self-Refine: Iterative refinement with self-feedback," in *Advances in Neural Information Processing Systems*, 2023.

[31] N. Shinn, B. Labash, and A. Gopinath, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems*, 2023.

[32] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[33] C. Mou, X. Wang, L. Xie, Y. Wu, J. Zhang, Z. Qi, Y. Shan, and X. Qie, "T2I-Adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

[34] Y. Guo, C. Yang, A. Rao, Z. Liang, Y. Wang, Y. Qiao, M. Agrawala, D. Lin, and B. Dai, "AnimateDiff: Animate your personalized text-to-image diffusion models without specific tuning," in *International Conference on Learning Representations*, 2024.

[35] T. Unterthiner, S. van Steenkiste, K. Kurach, R. Marinier, M. Michalski, and S. Gelly, "Towards accurate generative models of video: A new metric and challenges," arXiv preprint arXiv:1812.01717, 2018.

[36] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the International Conference on Machine Learning*, 2021.

[37] J. Hessel, A. Holtzman, M. Forbes, R. Le Bras, and Y. Choi, "CLIPScore: A reference-free evaluation metric for image captioning," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, 2021.

[38] Y. Liu, X. Cun, X. Liu, X. Wang, Y. Zhang, H. Chen, Y. Liu, T. Zeng, R. Chan, and Y. Shan, "EvalCrafter: Benchmarking and evaluating large video generation models," arXiv preprint arXiv:2310.11440, 2023.

[39] T.-H. Wu, L. Lian, J. E. Gonzalez, B. Li, and T. Darrell, "Self-correcting LLM-controlled diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. "'

References

ROHANSANJAYPOL

M240006CS_Updated_Report__Copy___Copy_.pdf

National Institute of Technology Sikkim

Document Details

Submission ID
trn:oid:::3117:596394123

Submission Date
Jun 1, 2026, 12:58 PM GMT+5:30

Download Date
Jun 1, 2026, 1:00 PM GMT+5:30

File Name
M240006CS_Updated_Report__Copy___Copy_.pdf

File Size
5.4 MB

78 Pages

20,382 Words

124,122 Characters

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Cited Text
- Small Matches (less than 8 words)
- Methods and Materials

Match Groups

89 Not Cited or Quoted 5%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
3% Publications
0% Submitted works (Student Papers)

Integrity Flags

1 Integrity Flag for Review

Replaced Characters
82 suspect characters on 21 pages
Letters are swapped with similar characters from another alphabet.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **89** Not Cited or Quoted 5%
  Matches with neither in-text citation nor quotation marks
- **0** Missing Quotations 0%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
3% Publications
0% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	arxiv.org 1%
2	Internet	www.coursehero.com <1%
3	Internet	docs.belle2.org <1%
4	Internet	nitsikkim.ac.in <1%
5	Internet	export.arxiv.org <1%
6	Internet	ethesis.nitrkl.ac.in <1%
7	Internet	www.frontiersin.org <1%
8	Internet	www.diva-portal.org <1%
9	Publication	Ton Duc Thang University <1%
10	Internet	digitalcollection.utem.edu.my <1%

11	Internet	
eprints.lancs.ac.uk		<1%
12	Internet	
gistperssanwa.web.app		<1%
13	Internet	
brage.bibsys.no		<1%
14	Internet	
escholarship.org		<1%
15	Internet	
www.preprints.org		<1%
16	Internet	
www.scribd.com		<1%
17	Internet	
gyan.iitg.ac.in		<1%
18	Internet	
cvru.ac.in		<1%
19	Internet	
www.ijimai.org		<1%
20	Publication	
"Advanced Intelligent Computing Technology and Applications", Springer Science...		<1%
21	Publication	
Jing (Selina) He, Mr. Shouling Ji, Yingshu Li, Yi Pan. "Wireless Ad Hoc and Sensor N...		<1%
22	Internet	
vesit.ves.ac.in		<1%
23	Publication	
Chaoyang Li, Xin Wang, Wenwu Zhu, Lingzhi Wang, Ning Hu, Qing Liao. "Customi...		<1%
24	Publication	
He, Xuehai. "Bridging the Gap Between Multimodal Foundation Models and Worl...		<1%

25 Publication

Mahendra Singh Sisodia, Venkat Prasad Padhy. "A novel Adaptive Loss Weighting... <1%

26 Internet

doczz.net <1%

27 Internet

www.cg.tuwien.ac.at <1%

28 Publication

"Pattern Recognition and Computer Vision", Springer Science and Business Media... <1%

29 Publication

Chen, Yunuo. "Physically Grounded World Modeling: Simulation, Generation, and ... <1%

30 Publication

Joanna Materzyńska, Josef Sivic, Eli Shechtman, Antonio Torralba, Richard Zhang, ... <1%

31 Internet

www.usenix.org <1%

32 Publication

"Proceedings of the First International Conference on Intelligent Computing and ... <1%

33 Publication

Ekin, Yiğit. "Controllable Diffusion-Based Visual Editing.", Bilkent Universitesi (Tur... <1%

34 Publication

Gosselin, Anthony. "Controllable Realistic Simulation and Video Generation for A... <1%

35 Publication

Huang, Hsin-Ping. "Generative Models for Novel Views and Video Synthesis", Univ... <1%

36 Publication

Paolo Ferro, Harinadh Vemanaboina, Chander Prakash. "Computational Techniqu... <1%

37 Publication

Songwei Ge, Seungjun Nah, Guilin Liu, Tyler Poon et al. "Preserve Your Own Corre... <1%

38 Internet

era.library.ualberta.ca <1%

39 Internet

theses.hal.science <1%

40 Internet

ualberta.scholaris.ca <1%

41 Internet

www.arxiv-vanity.com <1%

ROHANSANJAYPOL

M240006CS_Updated_Report__Copy___Copy_.pdf

National Institute of Technology Sikkim

Document Details

Submission ID
trn:oid:::3117:596394123

Submission Date
Jun 1, 2026, 12:58 PM GMT+5:30

Download Date
Jun 1, 2026, 1:06 PM GMT+5:30

File Name
M240006CS_Updated_Report__Copy___Copy_.pdf

File Size
5.4 MB

78 Pages

20,382 Words

124,122 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.